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Chapter 8

Profit, Loss and Discount

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Exercise 8(A)

Question 1(i)

10 articles are bought for ₹ 40 and are sold at ₹ 5 per article. The profit / loss made is:

1. 25% loss
2. 25% profit
3. 20% loss



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4. 20% profit

Answer

C.P. of 10 articles = ₹ 40

$$\text{C.P. of 1 article} = ₹ \frac{40}{10} = ₹ 4$$

S.P. of 1 article = ₹ 5

(∵ S.P. is greater than C.P., means article is sold at a profit.)

Profit = S.P. - C.P.

$$= ₹ 5 - ₹ 4 = ₹ 1$$

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P.}} \times 100 \%$$

$$= \frac{1}{4} \times 100 \%$$

$$= \frac{100}{4} \%$$

$$= 25 \%$$

Hence, option 2 is the correct option.**Question 1(ii)**

A table is sold at 80% of its cost price. The profit or loss as percent is :

1. 20% loss

2. 20% profit

3. 25% loss

4. 25% profit

Answer

Lets take the C.P. of table to be ₹ 100.

S.P. of table = ₹ 80 of cost price.

$$\text{S.P.} = ₹ \left(\frac{80}{100} \times 100 \right)$$

$$= ₹ \left(\frac{80}{100} \times 100 \right)$$

$$= ₹ 80$$

($\because$ C.P. is greater than S.P., means table is sold at a loss.)

Loss = C.P. - S.P.

$$= ₹ 100 - ₹ 80 = ₹ 20$$

$$\text{Loss\%} = \frac{\text{Loss}}{\text{C.P.}} \times 100\%$$

$$= \frac{20}{100} \times 100\%$$

$$= \frac{20}{100} \times 100\%$$

$$= 20\%$$

Hence, option 1 is the correct option.

Question 1(iii)

C.P. = ₹ 150 and loss = ₹ 50 $\Rightarrow$ the S.P. is:

1. ₹ 200
2. ₹ 100
3. ₹ 225
4. ₹ 75

Answer

Given:

C.P. = ₹ 150

Loss = ₹ 50

Loss = C.P. - S.P.

Putting the values, we get

$$\text{₹ } 50 = \text{₹ } 150 - \text{S.P.}$$

$$\text{S.P.} = \text{₹ } 150 - \text{₹ } 50$$

$$= \text{₹ } 100$$

Hence, option 2 is the correct option.

Question 1(iv)

C.P. = ₹ 150 and loss = 50% $\Rightarrow$ the S.P. is:

1. ₹ 200

2. ₹ 100

3. ₹ 225

4. ₹ 75

Answer

Given:

C.P. = ₹ 150

Loss % = 50%

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 50 = \frac{\text{Loss}}{150} \times 100$$

$$\Rightarrow \text{Loss} = \frac{50 \times 150}{100}$$

$$= \frac{7500}{100}$$

$$= 75$$

As we know:

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow 75 = 150 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 150 - 75$$

$$\Rightarrow \text{S.P.} = 75$$

Hence, option 4 is the correct option.

Question 1(v)

S.P. = ₹ 250 and profit = 25% $\Rightarrow$ the C.P. is:

1. ₹ 225

2. ₹ 275

3. ₹ 200

4. ₹ none of these

Answer

Given:

$$\text{S.P.} = ₹ 250$$

$$\text{Profit \%} = 25\%$$

$$\text{Let C.P.} = ₹x.$$

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 25 = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{25 \times x}{100}$$

$$= \frac{x}{4}$$

As we know:

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{4} = 250 - x$$

$$\Rightarrow 250 = \frac{x}{4} + x$$

$$\Rightarrow 250 = \frac{x}{4} + \frac{4x}{4}$$

$$\Rightarrow 250 = \frac{(x + 4x)}{4}$$

$$\Rightarrow 250 = \frac{5x}{4}$$

$$\Rightarrow x = \frac{4 \times 250}{5}$$

$$\Rightarrow x = \frac{1000}{5}$$

$$\Rightarrow x = 200$$

Hence, option 3 is the correct option.

Question 1(vi)

C.P. = ₹ 400 and overheads = ₹ 100. If loss = 10%; the S.P. is :

1. ₹ 360

2. ₹ 440

3. ₹ 450

4. ₹ 550

Answer**Given:**

C.P. = ₹ 400

Overheads = ₹ 100

Hence, total C.P. = C.P. + overheads

$$= ₹ 400 + ₹ 100 = ₹ 500$$

Loss % = 10%

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 10 = \frac{\text{Loss}}{500} \times 100$$

$$\Rightarrow \text{Loss} = \frac{10 \times 500}{100}$$

$$= \frac{5000}{100}$$

$$= 50$$

As we know:

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow 50 = 500 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 500 - 50$$

$$= 450$$

Hence, option 3 is the correct option.

Question 2

A fruit-seller buys oranges at 4 for ₹ 8 and sells them at 3 for ₹ 9. Find his profit percent.

Answer

$$\text{C.P. of 4 oranges} = ₹ 8$$

$$\text{C.P. of 1 orange} = ₹ \frac{8}{4} = ₹ 2$$

$$\text{S.P. of 3 oranges} = ₹ 9$$

$$\text{S.P. of 1 orange} = ₹ \frac{9}{3} = ₹ 3$$

($\because$ S.P. is greater than C.P., means oranges are sold at a profit.)

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$= ₹ 3 - ₹ 2 = ₹ 1$$

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100 \%$$

$$= \frac{1}{2} \times 100 \%$$

$$= \frac{100}{2} \%$$

$$= 50 \%$$

Hence, the profit percent = 50.

Question 3

A man buys a certain number of articles at 15 for ₹ 112.50 and sells them at 12 for ₹ 108. Find :

(i) his gain as percent;

(ii) the number of articles sold to make a profit of ₹ 75.

Answer

(i) C.P. of 15 articles = ₹ 112.50

$$\text{C.P. of 1 article} = ₹ \frac{112.50}{15} = ₹ 7.5$$

S.P. of 12 articles = ₹ 108

$$\text{S.P. of 1 article} = ₹ \frac{108}{12} = ₹ 9$$

($\because$ S.P. is greater than C.P., means article is sold at a profit.)

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$= ₹ 9 - ₹ 7.5 = ₹ 1.5$$

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100 \%$$

$$= \frac{1.5}{7.5} \times 100 \%$$

$$= \frac{150}{7.5} \%$$

$$= \frac{1500}{75} \%$$

$$= 20 \%$$

Hence, the profit percent = 20.

(ii) Profit on 1 article = ₹ 1.5

Lets suppose x articles are sold to make the profit of ₹ 75.

Profit on x articles = ₹ 75

No. of articles $\times$ Profit on 1 article = Total Profit

$$\Rightarrow x \times 1.5 = 75$$

$$\Rightarrow x = \frac{75}{1.5}$$

$$\Rightarrow x = \frac{750}{15}$$

$$\Rightarrow x = 50$$

Hence, 50 articles need to be sold to make a profit of ₹ 75.

Question 4

A boy buys an old bicycle for ₹ 162 and spends ₹ 18 on its repairs before selling the bicycle for ₹ 207. Find his gain or loss as percent.

Answer

C.P. of bicycle = ₹ 162

Amount spent on its repair = ₹ 18

Final C.P. = ₹ 162 + ₹ 18 = ₹ 180

S.P. of bicycle = ₹ 207

($\because$ S.P. is greater than C.P., means bicycle is sold at a gain.)

Gain = S.P. - C.P.

= ₹ 207 - ₹ 180

= ₹ 27

$$\text{Gain \%} = \frac{\text{Gain}}{\text{C.P.}} \times 100\%$$

$$= \frac{27}{180} \times 100\%$$

$$= \frac{2700}{180}\%$$
$$= 15\%$$

Hence, bicycle is sold at a gain of 15%.

Question 5

An article is bought from Jaipur for ₹ 4,800 and is sold in Delhi for ₹ 5,820. If ₹ 1,200 is spent on its transportation, etc., find the loss or the gain as percent.

Answer

Given:

C.P. of an article = ₹ 4,800

Cost spent on transportation, etc. = ₹ 1,200

Total C.P. = ₹ 4,800 + ₹ 1,200

= ₹ 6,000

S.P. of the article = ₹ 5,820

(∵ C.P. is greater than S.P., means article is sold at loss.)

Loss = C.P. - S.P.

= ₹ 6,000 - ₹ 5,820

= ₹ 180

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100\%$$

$$\begin{aligned} &= \frac{180}{6000} \times 100\% \\ &= \frac{18,000}{6000}\% \\ &= 3\% \end{aligned}$$

Hence, article is sold at a loss of 3%.

Question 6

Mohit sold a T.V. for ₹ 3,600, gaining one-sixth of its selling price. Find :

- (i) the gain.
- (ii) the cost price of the T.V.
- (iii) the gain percent.

Answer

(i) **Given:**

S.P. of T.V. = ₹ 3,600

Gain = one-sixth of its selling price

$$\begin{aligned} \text{Gain} &= ₹ \left(\frac{1}{6} \times 3,600 \right) \\ &= ₹ \frac{3,600}{6} \\ &= ₹ 600 \end{aligned}$$

Gain = ₹ 600

(ii) As we know that,

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

Putting the values, we get

$$₹ 600 = ₹ 3,600 - \text{C.P.}$$

$$\text{C.P.} = ₹ 3,600 - ₹ 600$$

$$= ₹ 3,000$$

The cost price of the T.V. = ₹ 3,000.

(iii)

$$\text{Gain \%} = \frac{\text{Gain}}{\text{C.P.}} \times 100\%$$

$$= \frac{600}{3000} \times 100\%$$

$$= \frac{1}{5} \times 100\%$$

$$= \frac{100}{5}\%$$

$$= 20\%$$

Hence, T.V. is sold at a gain of 20%.

Question 7

By selling a certain number of goods for ₹ 5,500, a shopkeeper loses equal to one-tenth of their selling price. Find :

- (i) the loss incurred
- (ii) the cost price of the goods
- (iii) the loss as percent.

Answer

(i) **Given:**

S.P. of a certain number of goods = ₹ 5,500

Loss = one-tenth of its selling price

$$= ₹ \frac{1}{10} \times 5,500$$

$$= ₹ \frac{5,500}{10}$$

$$= ₹ 550$$

Loss = ₹ 550

(ii) As we know that,

Loss = C.P. - S.P.

Putting the values, we get

$$₹ 550 = \text{C.P.} - ₹ 5,500$$

$$\text{C.P.} = ₹ 5,500 + ₹ 550$$

$$= ₹ 6,050$$

The cost price of the goods = ₹ 6,050.

(iii)

$$\begin{aligned}\text{Loss \%} &= \frac{\text{Loss}}{\text{C.P.}} \times 100\% \\ &= \frac{550}{6050} \times 100\% \\ &= \frac{55,000}{6050}\% \\ &= \frac{100}{11}\% \\ &= 9\frac{1}{11}\%\end{aligned}$$

Hence, T.V. is sold at a loss of $9\frac{1}{11}$.

Question 8

The selling price of a sofa set is $\frac{4}{5}$ times of its cost price. Find the gain or the loss as percent.

Answer

Given:

The S.P. of a sofa set = $\frac{4}{5}$ times of its C.P.

Let the cost price of sofa set is ₹100.

$$\text{S.P.} = ₹ \frac{4}{5} \times 100$$

$$= ₹ \frac{400}{5}$$

$$= ₹80$$

($\because$ C.P. is greater than S.P., means sofa is sold at a loss.)

Loss = C.P. - S.P.

$$= ₹100 - ₹80$$

$$= ₹20$$

$$\text{Loss\%} = \frac{\text{Loss}}{\text{C.P.}} \times 100\%$$

$$= \frac{20}{100} \times 100\%$$

$$= \frac{20}{\cancel{100}} \times \cancel{100}\%$$

$$= 20\%$$

Hence, Sofa is sold at a loss of 20%.

Question 9

The cost price of an article is $\frac{4}{5}$ times of its selling price. Find the loss or the gain as percent.

Answer

Given:

The C.P. of an article = $\frac{4}{5}$ times of its S.P.

Let the S.P. be ₹ 100.

$$\text{C.P.} = ₹ \frac{4}{5} \times 100$$

$$= ₹ \frac{400}{5}$$

$$= ₹ 80$$

($\because$ S.P. is greater than C.P., means article is sold at gain.)

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$= ₹ 100 - ₹ 80$$

$$= ₹ 20$$

$$\text{Gain}\% = \frac{\text{Gain}}{\text{C.P.}} \times 100\%$$

$$= \frac{20}{80} \times 100\%$$

$$= \frac{1}{4} \times 100\%$$

$$= \frac{100}{4}\%$$

$$= 25\%$$

Hence, Gain% = 25%

Question 10

The cost price of an article is 90% of its selling price. What is the profit or the loss as percent ?

Answer**Given:**

The C.P. of an article = 90% of its S.P.

Let the S.P. be ₹ 100.

$$\text{C.P.} = ₹ \frac{90}{100} \times 100$$

$$= ₹ \frac{90}{100} \times 100$$

$$= ₹ 90$$

($\because$ S.P. is greater than C.P., means article is sold at profit.)

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$= ₹ 100 - ₹ 90$$

$$= ₹ 10$$

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100\%$$

$$= \frac{10}{90} \times 100\%$$

$$= \frac{1}{9} \times 100\%$$

$$= \frac{100}{9}\%$$

$$= 11\frac{1}{9}\%$$

$$\text{Hence, Profit\%} = 11\frac{1}{9}\%$$

Question 11

The cost price of an article is 30 percent less than its selling price. Find the profit or the loss as percent.

Answer**Given:**

The C.P. of an article = 30% less than its S.P.

Let the S.P. be ₹ 100.

$$\text{C.P.} = \text{S.P.} - \frac{30}{100} \times \text{S.P.}$$

$$\text{C.P.} = ₹100 - \frac{30}{100} \times 100$$

$$= ₹100 - \frac{30}{100} \times 100$$

$$= ₹100 - 30$$

$$= ₹70$$

($\because$ S.P. is greater than C.P., means article is sold at profit.)

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$= ₹ 100 - ₹ 70$$

$$= ₹ 30$$

$$\begin{aligned}\text{Profit \%} &= \frac{\text{Profit}}{\text{C.P.}} \times 100\% \\&= \frac{30}{70} \times 100\% \\&= \frac{3,000}{70}\% \\&= \frac{300}{7}\% \\&= 42\frac{6}{7}\%\end{aligned}$$

$$\text{Hence, Profit\%} = 42\frac{6}{7}\%$$

Question 12

A shopkeeper bought 300 eggs at 80 paise each. 30 eggs were broken in transit and then he sold the remaining eggs at one rupee each. Find his gain or loss as percent.

Answer

C.P. of 1 egg = 80 paise

$$\begin{aligned}\text{C.P. of 300 eggs} &= 300 \times \frac{80}{100} \\&= 300 \times \frac{4}{5} \\&= \frac{1,200}{5} \\&= ₹ 240\end{aligned}$$

S.P. of 1 egg = ₹ 1

Total number of eggs = 300

Broken eggs = 30

Remaining eggs = $300 - 30 = 270$

S.P. of 270 egg = $270 \times ₹ 1 = ₹ 270$

($\because$ S.P. is greater than C.P., means eggs are sold at gain.)

Gain = S.P. - C.P.

= ₹ 270 - ₹ 240

= ₹ 30

$$\text{Gain}\% = \frac{\text{Gain}}{\text{C.P.}} \times 100\%$$

$$= \frac{30}{240} \times 100\%$$

$$= \frac{1}{8} \times 100\%$$

$$= \frac{100}{8}\%$$

$$= 12.5\%$$

Hence, Gain% = 12.5%

Question 13

By selling an article for ₹ 900, a man gains 20%. Find his cost price and the gain.

Answer

Given:

$$\text{S.P.} = ₹ 900$$

$$\text{Gain \%} = 20\%$$

$$\text{Let C.P.} = ₹ x.$$

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 20 = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{20 \times x}{100}$$

$$= \frac{x}{5}$$

As we know:

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{5} = 900 - x$$

$$\Rightarrow 900 = \frac{x}{5} + x$$

$$\Rightarrow 900 = \frac{x}{5} + \frac{5x}{5}$$

$$\Rightarrow 900 = \frac{(x + 5x)}{5}$$

$$\Rightarrow 900 = \frac{6x}{5}$$

$$\Rightarrow x = \frac{5 \times 900}{6}$$

$$\Rightarrow x = \frac{4500}{6}$$

$$\Rightarrow x = 750$$

$$\text{Gain} = \frac{x}{5}$$

$$\text{Gain} = \frac{750}{5} = 150$$

Hence, C.P. = ₹ 750 and gain = ₹ 150.

Question 14

By selling an article for ₹ 704, a person loses 12%. Find his cost price and the loss.

Answer

Given:

$$\text{S.P.} = ₹ 704$$

$$\text{Loss \%} = 12\%$$

Let C.P. = ₹ x .

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 12 = \frac{\text{Loss}}{x} \times 100$$

$$\Rightarrow \text{Loss} = \frac{12 \times x}{100}$$

$$= \frac{3x}{25}$$

As we know:

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{3x}{25} = x - 704$$

$$\Rightarrow 704 = x - \frac{3x}{25}$$

$$\Rightarrow 704 = \frac{25x}{25} - \frac{3x}{25}$$

$$\Rightarrow 704 = \frac{(25x - 3x)}{25}$$

$$\Rightarrow 704 = \frac{22x}{25}$$

$$\Rightarrow x = \frac{25 \times 704}{22}$$

$$\Rightarrow x = \frac{17,600}{22}$$

$$\Rightarrow x = ₹800$$

$$\text{Loss} = \frac{3x}{25}$$

$$\text{Gain} = \frac{3 \times 800}{25}$$

$$\text{Gain} = \frac{2400}{25} = 96$$

Hence, C.P. = ₹ 800 and loss = ₹ 96.

Question 15

Find the selling price :

(i) C.P. = ₹ 352, overheads = ₹ 28 and profit = 20%.

(ii) C.P. = ₹ 576, overheads = ₹ 44 and loss = 16%.

Answer

(i) **Given:**

C.P. = ₹ 352

Overheads = ₹ 28

Hence, total C.P. = C.P. + overheads

= ₹ 352 + ₹ 28 = ₹ 380

Profit % = 20%

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 20 = \frac{\text{Profit}}{380} \times 100$$

$$\Rightarrow \text{Profit} = \frac{20 \times 380}{100}$$

$$= \frac{7600}{100}$$

$$= 76$$

As we know:

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow 76 = \text{S.P.} - 380$$

$$= 380 + 76$$

$$= 456$$

Hence, S.P. = ₹ 456.

(ii) **Given:**

$$\text{C.P.} = ₹ 576$$

$$\text{Overheads} = ₹ 44$$

Hence, total C.P. = C.P. + overheads

$$= ₹ 576 + ₹ 44 = ₹ 620$$

$$\text{Loss \%} = 16\%$$

Let the C.P. be ₹ x .

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 16 = \frac{\text{Loss}}{620} \times 100$$

$$\Rightarrow \text{Loss} = \frac{16 \times 620}{100}$$

$$= \frac{9,920}{100}$$

$$= 99.2$$

As we know:

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow 99.2 = 620 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 620 - 99.2$$

$$= 520.80$$

Hence, S.P. = ₹ 520.80.

Question 16

If John sells his bicycle for ₹ 637, he will suffer a loss of 9%. For how much should it be sold if he desires a profit of 5% ?

Answer

Given:

S.P. of bicycle = ₹ 637

Loss % = 9%

Let C.P. = ₹ x .

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 9 = \frac{\text{Loss}}{x} \times 100$$

$$\Rightarrow \text{Loss} = \frac{9 \times x}{100}$$

$$= \frac{9x}{100}$$

As we know:

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{9x}{100} = x - 637$$

$$\Rightarrow 637 = x - \frac{9x}{100}$$

$$\Rightarrow 637 = \frac{100x}{100} - \frac{9x}{100}$$

$$\Rightarrow 637 = \frac{(100x - 9x)}{100}$$

$$\Rightarrow 637 = \frac{91x}{100}$$

$$\Rightarrow x = \frac{100 \times 637}{91}$$

$$\Rightarrow x = \frac{63,700}{91}$$

$$\Rightarrow x = 700$$

Hence, C.P. = ₹ 700

Profit % = 5 %

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 5 = \frac{\text{Profit}}{700} \times 100$$

$$\Rightarrow \text{Profit} = \frac{5 \times 700}{100}$$

$$= \frac{3500}{100}$$

$$= 35$$

And,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

Putting all the values, we get

$$\Rightarrow 35 = \text{S.P.} - 700$$

$$\Rightarrow \text{S.P.} = 35 + 700$$

$$= 735$$

John should sell his bicycle for ₹ 735 to make a profit of 5%.

Question 17

A man sells a radio set for ₹ 605 and gains 10%. At what price should he sell another radio of the same kind in order to gain 16% ?

Answer

Given:

S.P. of radio set = ₹ 605

Gain % = 10 %

Let C.P. = ₹ x .

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 10 = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{10 \times x}{100}$$

$$= \frac{10x}{100}$$

$$= \frac{x}{10}$$

As we know:

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{10} = 605 - x$$

$$\Rightarrow 605 = \frac{x}{10} + x$$

$$\Rightarrow 605 = \frac{10x}{10} + \frac{x}{10}$$

$$\Rightarrow 605 = \frac{(10x + x)}{10}$$

$$\Rightarrow 605 = \frac{11x}{10}$$

$$\Rightarrow x = \frac{10 \times 605}{11}$$

$$\Rightarrow x = \frac{6,050}{11}$$

$$\Rightarrow x = 550$$

Hence, C.P. = ₹ 550

Profit % = 16 %

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 16 = \frac{\text{Profit}}{550} \times 100$$

$$\Rightarrow \text{Profit} = \frac{16 \times 550}{100}$$

$$= \frac{8800}{100}$$

$$= 88$$

And,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

Putting all the values, we get

$$\Rightarrow 88 = \text{S.P.} - 550$$

$$\Rightarrow \text{S.P.} = 88 + 550$$

$$= 638$$

The man should sell the other radio for ₹ 638 in order to make a profit of 16%.

Question 18

By selling a sofa set for ₹ 2,500, the shopkeeper loses 20%. Find his loss percent or profit percent, if he sells the same sofa set for ₹ 3,150.

Answer

Given:

S.P. of sofa set = ₹ 2,500

Loss % = 20 %

Let C.P. = ₹ x .

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 20 = \frac{\text{Loss}}{x} \times 100$$

$$\Rightarrow \text{Loss} = \frac{20 \times x}{100}$$

$$= \frac{x}{5}$$

As we know:

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{x}{5} = x - 2,500$$

$$\Rightarrow 2,500 = x - \frac{x}{5}$$

$$\Rightarrow 2,500 = \frac{5x}{5} - \frac{x}{5}$$

$$\Rightarrow 2,500 = \frac{(5x - x)}{5}$$

$$\Rightarrow 2,500 = \frac{4x}{5}$$

$$\Rightarrow x = \frac{2,500 \times 5}{4}$$

$$\Rightarrow x = \frac{12,500}{4}$$

$$\Rightarrow x = 3,125$$

When C.P. = ₹ 3,125

New S.P. = ₹ 3,150

($\because$ When S.P. is greater than C.P., means sofa set will be sold at a profit.)

Profit = S.P. - C.P.

$$= ₹ 3,150 - ₹ 3,125$$

$$= ₹ 25$$

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100\%$$

$$= \frac{25}{3125} \times 100\%$$

$$= \frac{1}{125} \times 100\%$$

$$= \frac{100}{125}\%$$
$$= 0.8\%$$

If the man sells the same sofa set for ₹ 3,150, profit % = 0.8%.

Exercise 8(B)

Question 1(i)

Identical pens are bought at 10 for ₹ 80. If these pens are sold at 25% profit; the S.P. per pen is :

1. ₹ 6
2. ₹ 10
3. ₹ 12
4. ₹ 8

Answer

C.P. of 10 pens = ₹ 80

$$\text{C.P. of 1 pen} = ₹ \frac{80}{10} = ₹ 8$$

Profit = 25%

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow 25 = \frac{\text{Profit}}{8} \times 100$$

$$\Rightarrow \text{Profit} = \frac{25 \times 8}{100}$$

$$= \frac{200}{100}$$

$$= 2$$

As we know,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

Putting the values, we get

$$\Rightarrow 2 = \text{S.P.} - 8$$

$$\Rightarrow \text{S.P.} = 2 + 8$$

$$\Rightarrow \text{S.P.} = 10$$

Hence, option 2 is the correct option.

Question 1(ii)

If C.P. of 20 identical articles is same as S.P. of 25 articles. The profit or the loss as percent is :

1. 25% loss
2. 25% profit
3. 20% loss
4. 20% profit

Answer

C.P. of 20 articles = S.P. of 25 articles

Let the C.P. of 20 articles be ₹ 100.

$$\text{The C.P. of 1 article} = ₹ \frac{100}{20} = ₹ 5$$

The S.P. of 25 articles = The C.P. of 20 articles = ₹ 100

$$\text{The S.P. of 1 article} = ₹ \frac{100}{25} = ₹ 4$$

($\because$ C.P. is greater than S.P., means articles are sold at a loss.)

Loss = C.P. - S.P.

$$= ₹ (5 - 4) = ₹ 1$$

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100\%$$

$$= \frac{1}{5} \times 100\%$$

$$= \frac{100}{5}\%$$

$$= 20\%$$

Hence, option 3 is the correct option.

Question 1(iii)

The marked price of an article is ₹ 500 and is sold for ₹ 400; the discount given is :

1. 10%
2. 20%
3. 25%
4. none of these

Answer

Given:

M.P. of an article = ₹ 500

S.P. of article = ₹ 400

Discount = M.P. - S.P.

$$= ₹ 500 - ₹ 400$$

$$= ₹ 100$$

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100\%$$

$$= \frac{100}{500} \times 100\%$$

$$= \frac{1}{5} \times 100\%$$

$$= \frac{100}{5}\%$$

$$= 20\%$$

Hence, option 2 is the correct option.

Question 1(iv)

An article is marked at ₹ 800 and is sold at 20% discount. Its S.P. is :

1. ₹ 160

2. ₹ 960

3. ₹ 640

4. ₹ 780

Answer

Given:

M.P. = ₹ 800

Discount % = 20%

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 20 = \frac{\text{Discount}}{800} \times 100$$

$$\Rightarrow \text{Discount} = \frac{20 \times 800}{100}$$

$$= \frac{16000}{100}$$

$$= 160$$

And we know,

$$\text{Discount} = \text{M.P.} - \text{S.P}$$

Putting the values, we get

$$160 = 800 - \text{S.P}$$

$$\Rightarrow \text{S.P.} = 800 - 160$$

$$\Rightarrow \text{S.P.} = 640$$

Hence, option 3 is the correct option.

Question 1(v)

By selling 10 articles, a man gains equal to cost price of 2 articles, the profit made is :

1. 40%

2. 20%

3. 25%

4. 50%

Answer

Let the C.P. of 1 article be ₹ 100.

So, the C.P. of 2 article = ₹ 100 x 2 = ₹ 200

Similarly, the C.P. of 10 article = ₹ 100 x 10 = ₹ 1000

Gain = Cost price of 2 articles = ₹ 200

As we know,

$$\begin{aligned}\text{Profit\%} &= \frac{\text{Profit}}{\text{C.P.}} \times 100\% \\ &= \frac{200}{1000} \times 100\% \\ &= \frac{1}{5} \times 100\% \\ &= \frac{100}{5}\% \\ &= 20\%\end{aligned}$$

Hence, option 2 is the correct option.

Question 2

A fruit-seller sells 4 oranges for ₹ 3, gaining 50%. Find :

(i) C.P. of 4 oranges.

(ii) C.P. of one orange.

(iii) S.P. of one orange

(iv) profit made by selling one orange

(v) number of oranges need to be bought and sold in order to gain ₹ 24.

Answer

(i) **Given:**

S.P. of 4 oranges = ₹ 3

S.P. of 1 orange = ₹ $\frac{3}{4}$

Gain = 50%

Let C.P. = ₹ x .

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 50 = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{50 \times x}{100}$$

$$= \frac{x}{2}$$

As we know:

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{2} = \frac{3}{4} - x$$

$$\Rightarrow \frac{3}{4} = \frac{x}{2} + x$$

$$\Rightarrow \frac{3}{4} = \frac{x}{2} + \frac{2x}{2}$$

$$\Rightarrow \frac{3}{4} = \frac{(x + 2x)}{2}$$

$$\Rightarrow \frac{3}{4} = \frac{3x}{2}$$

$$\Rightarrow x = \frac{3 \times 2}{4 \times 3}$$

$$\Rightarrow x = \frac{6}{12}$$

$$\Rightarrow x = \frac{1}{2}$$

$$\text{C.P. of 1 orange} = ₹ \frac{1}{2}$$

$$\text{C.P. of 4 oranges} = ₹ 4 \times \frac{1}{2}$$

$$= ₹ \frac{4}{2}$$

$$= ₹ 2$$

Hence, C.P. of 4 oranges = ₹ 2.

$$(ii) \text{ C.P. of 1 orange} = ₹ \frac{1}{2} = ₹ 0.5$$

$$(iii) \text{ S.P. of 1 orange} = ₹ \frac{3}{4} = ₹ 0.75$$

$$(iv) \text{ Profit} = \text{S.P.} - \text{C.P.}$$

$$= ₹ 0.75 - ₹ 0.50$$

$$= ₹ 0.25$$

Hence, Profit = ₹ 0.25.

$$(v) \text{ Gain on 1 orange} = ₹ 0.25$$

Let number of oranges needed to gain ₹ 24 be x

$$\text{Gain on } x \text{ orange} = ₹ 0.25 \times x = ₹ 24$$

$$\Rightarrow x = \frac{24}{0.25}$$

$$\Rightarrow x = \frac{2400}{25}$$

$$\Rightarrow x = 96$$

Hence, 96 oranges need to be bought and sold in order to gain ₹ 24.

Question 3

A man sells 12 articles for ₹ 80 gaining $33\frac{1}{3}\%$. Find the number of articles bought by the man for ₹ 90.

Answer

Given:

S.P. of 12 articles = ₹ 80

$$\text{S.P. of 1 article} = ₹ \frac{80}{12} = ₹ \frac{20}{3}$$

$$\text{Gain} = 33\frac{1}{3} = \frac{100}{3}$$

Let C.P. of 1 article = ₹ x .

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow \frac{100}{3} = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{100 \times x}{3 \times 100}$$

$$= \frac{\cancel{100} \times x}{3 \times \cancel{100}}$$

$$= \frac{x}{3}$$

As we know:

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{3} = \frac{20}{3} - x$$

$$\Rightarrow \frac{20}{3} = \frac{x}{3} + x$$

$$\Rightarrow \frac{20}{3} = \frac{x}{3} + \frac{3x}{3}$$

$$\Rightarrow \frac{20}{3} = \frac{(x + 3x)}{3}$$

$$\Rightarrow \frac{20}{3} = \frac{4x}{3}$$

$$\Rightarrow x = \frac{20 \times 3}{4 \times 3}$$

$$\Rightarrow x = \frac{60}{12}$$

$$\Rightarrow x = 5$$

C.P. of 1 article = ₹ 5

Let number of articles bought by the man for ₹ 90 be y

C.P. of y articles = ₹ 5 $\times$ y = ₹ 90

$$\Rightarrow y = \frac{90}{5}$$

$$\Rightarrow y = 18$$

Hence, 18 articles are bought by the man for ₹ 90.

Question 4

The selling price of 15 articles is equal to the cost price of 12 articles. Find the gain or loss as percent.

Answer

Let the C.P. of 1 article be ₹ 100.

The C.P. of 12 articles = ₹ 100 x 12 = ₹ 1200

The S.P. of 15 articles = the C.P. of 12 articles

The S.P. of 15 articles = ₹ 1200

The S.P. of 1 article = ₹ $\frac{1200}{15}$ = ₹ 80

($\because$ C.P. is greater than S.P., means article is sold at a loss.)

Loss = C.P. - S.P.

= ₹ 100 - ₹ 80

= ₹ 20

Loss % = $\frac{\text{Loss}}{\text{C.P.}} \times 100\%$

= $\frac{20}{100} \times 100\%$

= $\frac{20}{100} \times 100\%$

= 20%

Hence, loss % = 20%.

Question 5

A shopkeeper bought rice worth ₹ 4,500. He sold one-third of it at 10% profit. If he desires a profit of 12% on the whole, find :

(i) the selling price of the rest of the rice.

(ii) the percentage profit on the rest of the rice.

Answer

(i) **Given:**

C.P. of whole rice = ₹ 4,500

Profit desired on the whole = 12 %

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P}} \times 100$$

Putting the values, we get

$$12 = \frac{\text{Profit}}{4,500} \times 100$$

$$\Rightarrow \text{Profit} = \frac{12 \times 4,500}{100}$$

$$= \frac{54,000}{100}$$

$$= 540$$

As we know,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

Putting the values, we get

$$540 = \text{S.P.} - 4,500$$

$$\Rightarrow \text{S.P.} = 540 + 4,500$$

$$= 5,040$$

$$\text{C.P. of } \frac{1}{3} \text{ rice} = ₹ \frac{1}{3} \times 4,500$$

$$= ₹ \frac{4,500}{3}$$

$$= ₹ 1,500$$

Gain on it = 10%

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$10 = \frac{\text{Profit}}{1,500} \times 100$$

$$\Rightarrow \text{Profit} = \frac{10 \times 1,500}{100}$$

$$= \frac{15,000}{100}$$

$$= 150$$

As we know,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

Putting the values, we get

$$150 = \text{S.P.} - 1,500$$

$$\Rightarrow \text{S.P.} = 150 + 1,500$$

$$= 1,650$$

The S.P. of the rest of the rice = ₹ 5,040 - ₹ 1,650 = ₹ 3,390

The selling price of the rest of the rice = ₹ 3,390.

(ii) C.P. of the rest of the rice = ₹ 4,500 - ₹ 1,500 = ₹ 3,000

Profit on the rest of the rice = Remaining S.P. - Remaining C.P.

$$= ₹ 3,390 - ₹ 3,000$$

$$= ₹ 390$$

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P.}} \times 100\%$$

$$= \frac{390}{3000} \times 100\%$$

$$= \frac{13}{100} \times 100\%$$

$$= \frac{1300}{100}\%$$

$$= 13\%$$

The profit percentage of rest of the rice = 13%.

Question 6

Mohan bought a certain number of notebooks for ₹ 600. He sold $\frac{1}{4}$ of them at 5 percent loss. At what price should he sell the remaining notebooks so as to gain 10% on the whole ?

Answer

Given:

C.P. of notebooks = ₹ 600

Gain desired on the whole = 10 %

$$\text{Profit\%} = \frac{\text{profit}}{\text{C.P}} \times 100\%$$

Putting the values, we get

$$10 = \frac{\text{Profit}}{600} \times 100$$

$$\Rightarrow \text{Profit} = \frac{10 \times 600}{100}$$

$$= \frac{6000}{100}$$

$$= 60$$

As we know,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

Putting the values, we get

$$60 = \text{S.P.} - 600$$

$$\Rightarrow \text{S.P.} = 60 + 600$$

$$= 660$$

$$\text{C.P. of } \frac{1}{4} \text{ number of notebooks} = ₹ \frac{1}{4} \times 600$$

$$= ₹ \frac{600}{4}$$

$$= ₹ 150$$

$$\text{Loss of } \frac{1}{4} \text{ number of notebooks} = 5\%$$

$$\text{Loss}\% = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$5 = \frac{\text{Loss}}{150} \times 100$$

$$\Rightarrow \text{Loss} = \frac{5 \times 150}{100}$$

$$= \frac{750}{100}$$

$$= 7.5$$

As we know,

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

Putting the values, we get

$$7.5 = 150 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 150 - 7.5$$

$$= 142.5$$

The S.P. of rest of notebooks = ₹ 660 - ₹ 142.5 = ₹ 517.5

The selling price of the rest of the notebooks = ₹ 517.50.

Question 7

Raju sells a watch at 5% profit. Had he sold it for ₹ 24 more he would have gained 11%. Find the cost price of the watch.

Answer

Let C.P. of the watch be ₹ x .

Profit % = 5 %

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P}} \times 100$$

Putting the values, we get

$$5 = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{5 \times x}{100}$$

$$= \frac{x}{20}$$

As we know,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

Putting the values, we get

$$\frac{x}{20} = \text{S.P.} - x$$

$$\Rightarrow \text{S.P.} = \frac{x}{20} + x$$

$$= \frac{x}{20} + \frac{20x}{20}$$

$$= \frac{(x + 20x)}{20}$$

$$= \frac{21x}{20}$$

Given,

If S.P. was 24 more gain would have been 11%

$$\text{New S.P.} = \frac{21x}{20} + 24$$

$$\text{New Profit} = \text{New S.P.} - \text{C.P.}$$

$$= \frac{21x}{20} + 24 - x$$

$$= \frac{21x + 480 - 20x}{20}$$

$$= \frac{x + 480}{20}$$

New Profit % = 11 %

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P}} \times 100$$

$$\Rightarrow 11 = \frac{\frac{x + 480}{20}}{x} \times 100$$

$$\Rightarrow 11 = \frac{x + 480}{20x} \times 100$$

$$\Rightarrow 11 = \frac{x + 480}{x} \times 5$$

$$\Rightarrow 11x = 5x + 2400$$

$$\Rightarrow 11x - 5x = 2400$$

$$\Rightarrow 6x = 2400$$

$$\Rightarrow x = \frac{2400}{6}$$

$$\Rightarrow x = 400$$

Hence, cost price of watch = ₹ 400.

Question 8

A wrist watch is available at a discount of 9%. If the list price of the watch is ₹ 1,400, find the discount given and the selling price of the watch.

Answer

Given:

M.P. of the watch = ₹ 1,400

Discount = 9%

$$\text{Discount}\% = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 9 = \frac{\text{Discount}}{1400} \times 100$$

$$\Rightarrow \text{Discount} = \frac{9 \times 1,400}{100}$$

$$= \frac{12,600}{100}$$

$$= ₹126$$

As we know,

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\Rightarrow 126 = 1,400 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 1,400 - 126$$

$$= ₹1,274$$

Hence, discount = ₹ 126 and S.P. = ₹ 1,274

Question 9

A shopkeeper sells an article for ₹ 248.50 after allowing a discount of 10% on its list price. Find the list price of the article.

Answer

Given:

S.P. of an article = ₹ 248.50

Discount = 10%

Let the M.P. of the article be ₹ x .

As we know,

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Discount}}{x} \times 100$$

$$\Rightarrow \text{Discount} = \frac{10 \times x}{100}$$

$$= \frac{10x}{100}$$

$$= \frac{x}{10}$$

And

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\Rightarrow \frac{x}{10} = x - 248.50$$

$$\Rightarrow 248.50 = x - \frac{x}{10}$$

$$\Rightarrow 248.50 = \frac{10x}{10} - \frac{x}{10}$$

$$\Rightarrow 248.50 = \frac{(10x - x)}{10}$$

$$\Rightarrow 248.50 = \frac{9x}{10}$$

$$\Rightarrow x = \frac{10 \times 248.50}{9}$$

$$\Rightarrow x = \frac{2485}{9}$$

$$\Rightarrow x = ₹276.11$$

Hence, the list price = ₹ 276.11

Question 10

A shopkeeper buys an article for ₹ 450. He marks it at 20% above the cost price. Find :

- (i) the marked price of the article.
- (ii) the selling price, if he sells the article at 10 percent discount.
- (iii) the percentage discount given by him, if he sells the article for ₹ 496.80.

Answer

(i) **Given:**

C.P. of an article = ₹ 450.

M.P. of the article = 20% above the C.P.

= C.P. + 20% of C.P.

$$= ₹ 450 + \frac{20}{100} \times 450$$

$$= ₹ 450 + \frac{1}{5} \times 450$$

$$= ₹ 450 + \frac{450}{5}$$

$$= ₹ 450 + 90$$

$$= ₹ 540$$

Hence, M.P. of the article = ₹ 540.

(ii) M.P. of the article = ₹ 540

Discount = 10 %

As we know,

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Discount}}{540} \times 100$$

$$\Rightarrow \text{Discount} = \frac{10 \times 540}{100}$$

$$= \frac{5400}{100}$$

$$= 54$$

And

$$\text{Discount} = \text{M.P.} - \text{S.P}$$

$$\Rightarrow 54 = 540 - \text{S.P}$$

$$\Rightarrow \text{S.P} = 540 - 54$$

$$\Rightarrow \text{S.P} = ₹486$$

Hence, S.P. of the article = ₹ 486.

(iii) When M.P. of the article = ₹ 540

S.P. of the article = ₹ 496.80

As we know ,

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\text{Discount} = 540 - 496.80$$

$$\text{Discount} = ₹43.2$$

And,

$$\text{Discount}\% = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\begin{aligned}\Rightarrow \text{Discount}\% &= \frac{43.2}{540} \times 100 \\ &= \frac{4320}{540}\% \\ &= 8\%\end{aligned}$$

If S.P. of the article is ₹ 496.80, then the percentage discount is 8%.

Question 11

The list price of an article is ₹ 800 and is available at a discount of 15 percent.

Find :

(i) the selling price of the article;

(ii) the cost price of the article if a profit of $13\frac{1}{3}\%$ is made on selling it.

Answer

(i) **Given:**

M.P. of an article = ₹ 800

Discount of the article = 15%

$$\text{Discount}\% = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 15 = \frac{\text{Discount}}{800} \times 100$$

$$\Rightarrow \text{Discount} = \frac{15 \times 800}{100}$$

$$= \frac{12000}{100}$$

$$= 120$$

And,

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\Rightarrow 120 = 800 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 800 - 120$$

$$\Rightarrow \text{S.P.} = 680 \text{ Hence, S.P. of the article} = ₹ 680.$$

(ii) S.P. of the article = ₹ 680

$$\text{Profit} = 13\frac{1}{3} = \frac{40}{3}$$

Let the C.P. be ₹ x .

As we know,

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow \frac{40}{3} = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{40 \times x}{3 \times 100}$$

$$= \frac{40x}{300}$$

$$= \frac{2x}{15}$$

And,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{2x}{15} = 680 - x$$

$$\Rightarrow \frac{2x}{15} + x = 680$$

$$\Rightarrow \frac{2x}{15} + \frac{15x}{15} = 680$$

$$\Rightarrow \frac{(2x + 15x)}{15} = 680$$

$$\Rightarrow \frac{17x}{15} = 680$$

$$\Rightarrow x = \frac{680 \times 15}{17}$$

$$\Rightarrow x = \frac{10,200}{17}$$

$$\Rightarrow x = 600$$

Hence, C.P. of the article = ₹ 600.

Question 12

An article is marked at ₹ 2,250. By selling it at a discount of 12%, the dealer makes a profit of 10%. Find :

(i) the selling price of the article.

(ii) the cost price of the article for the dealer.

Answer

(i) **Given:**

M.P. of an article = ₹ 2,250

Discount % = 12 %

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 12 = \frac{\text{Discount}}{2250} \times 100$$

$$\Rightarrow \text{Discount} = \frac{12 \times 2250}{100}$$

$$= \frac{27,000}{100}$$

$$= 270$$

Discount = M.P. - S.P.

$$\Rightarrow 270 = 2,250 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 2,250 - 270$$

$$\Rightarrow \text{S.P.} = 1,980$$

The S.P. of the article = ₹ 1,980.

(ii) S.P. of the article = ₹ 1,980

Profit of the article = 10%

Let the C.P. of the article be ₹ x .

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{10 \times x}{100}$$

$$= \frac{10x}{100}$$

$$= \frac{x}{10}$$

and,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{10} = 1980 - x$$

$$\Rightarrow \frac{x}{10} + x = 1980$$

$$\Rightarrow \frac{x}{10} + \frac{10x}{10} = 1980$$

$$\Rightarrow \frac{(x + 10x)}{10} = 1980$$

$$\Rightarrow \frac{11x}{10} = 1980$$

$$\Rightarrow x = \frac{1980 \times 10}{11}$$

$$\Rightarrow x = \frac{19800}{11}$$

$$\Rightarrow x = 1800$$

The cost price of the article = ₹ 1,800

Question 13

By selling an article at 20% discount, a shopkeeper gains 25%. If the selling price of the article is ₹ 1,440, find :

- (i) the marked price of the article.
- (ii) the cost price of the article.

Answer

(i) **Given:**

Discount on the article = 20%

S.P. of the article = ₹ 1,440

Let the M.P. of the article be ₹ x .

As we know,

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 20 = \frac{\text{Discount}}{x} \times 100$$

$$\Rightarrow \text{Discount} = \frac{20 \times x}{100}$$

$$\Rightarrow \text{Discount} = \frac{20x}{100}$$

$$\Rightarrow \text{Discount} = \frac{x}{5}$$

And,

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\Rightarrow \frac{x}{5} = x - 1440$$

$$\Rightarrow 1440 = x - \frac{x}{5}$$

$$\Rightarrow 1440 = \frac{5x}{5} - \frac{x}{5}$$

$$\Rightarrow 1440 = \frac{(5x - x)}{5}$$

$$\Rightarrow 1440 = \frac{4x}{5}$$

$$\Rightarrow x = \frac{5 \times 1,440}{4}$$

$$\Rightarrow x = \frac{7,200}{4}$$

$$\Rightarrow x = 1,800$$

Hence, M.P. of the article = ₹ 1,800

(ii) S.P. of the article = ₹ 1,440

Gain% of the article = 25%

Let the C.P. be ₹ y .

$$\text{Gain \%} = \frac{\text{Gain}}{\text{C.P.}} \times 100$$

$$\Rightarrow 25 = \frac{\text{Gain}}{y} \times 100$$

$$\Rightarrow \text{Gain} = \frac{25 \times y}{100}$$

$$\Rightarrow \text{Gain} = \frac{25y}{100}$$

$$\Rightarrow \text{Gain} = \frac{y}{4}$$

And,

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{y}{4} = 1,440 - y$$

$$\Rightarrow \frac{y}{4} + y = 1,440$$

$$\Rightarrow \frac{y}{4} + \frac{4y}{4} = 1,440$$

$$\Rightarrow \frac{(y + 4y)}{4} = 1,440$$

$$\Rightarrow \frac{5y}{4} = 1,440$$

$$\Rightarrow y = \frac{1,440 \times 4}{5}$$

$$\Rightarrow y = \frac{5,760}{5}$$

$$\Rightarrow y = 1,152$$

Hence, C.P. of the article = ₹ 1,152

Question 14

A shopkeeper marks his goods at 30 percent above the cost price and then gives a discount of 10 percent. Find his gain percent.

Answer

Given:

M.P. of the goods = 30% above the C.P.

Let the C.P. be ₹ 100.

M.P. = C.P. + 30% of C.P.

$$= 100 + \frac{30}{100} \times 100$$

$$= 100 + \frac{30}{100} \times 100$$

$$= 100 + 30$$

$$= 130$$

So, M.P. of the goods = ₹ 130

Discount % = 10%

$$\text{Discount}\% = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Discount}}{130} \times 100$$

$$\Rightarrow \text{Discount} = \frac{10 \times 130}{100}$$

$$= \frac{1300}{100}$$

$$= 13$$

Now,

Discount = M.P. - S.P.

$$\Rightarrow 13 = 130 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 130 - 13$$

$$= 117$$

Hence, S.P. of the goods = ₹ 117

C.P. of the goods = ₹ 100

($\because$ S.P. is greater than C.P., means goods are sold at a gain.)

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \text{Gain} = 117 - 100$$

$$\Rightarrow \text{Gain} = 17$$

And

$$\text{Gain}\% = \frac{\text{Gain}}{\text{C.P.}} \times 100\%$$

$$\Rightarrow \text{Gain}\% = \frac{17}{100} \times 100\%$$

$$= \frac{17}{100} \times 100\%$$

$$= 17\%$$

Hence, Gain percent = 17%

Question 15

A ready-made garments shop in Delhi allows 20 percent discount on its garments and still makes a profit of 20 percent. Find the marked price of a dress which is bought by the shopkeeper for ₹ 400.

Answer

Given:

C.P. of a dress = ₹ 400

Profit of the dress = 20%

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow 20 = \frac{\text{Profit}}{400} \times 100$$

$$\Rightarrow \text{Profit} = \frac{20 \times 400}{100}$$

$$= \frac{8000}{100}$$

$$= 80$$

And,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow 80 = \text{S.P.} - 400$$

$$\Rightarrow \text{S.P.} = 80 + 400$$

$$= 480$$

$$\text{S.P. of the dress} = ₹ 480$$

$$\text{Discount on the dress} = 20\%$$

$$\text{Let the M.P. of the dress} = ₹ x.$$

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\Rightarrow \text{Discount} = x - 480$$

And,

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 20 = \frac{(x - 480)}{x} \times 100$$

$$\Rightarrow 20 \times x = (x - 480) \times 100$$

$$\Rightarrow 20x = 100x - 48000$$

$$\Rightarrow 48000 = 100x - 20x$$

$$\Rightarrow 48000 = 80x$$

$$\Rightarrow x = \frac{48000}{80}$$

$$\Rightarrow x = 600$$

Hence, M.P. of the dress = ₹ 600.

Question 16

At 12% discount, the selling price of a pen is ₹ 13.20. Find its marked price.

Also find the new selling price of the pen, if it is sold at 5% discount.

Answer

Given:

S.P. of the pen = ₹ 13.20

Discount % = 12%

Let the M.P. of the pen = ₹ x .

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\Rightarrow \text{Discount} = x - 13.20$$

And,

$$\text{Discount}\% = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 12 = \frac{(x - 13.20)}{x} \times 100$$

$$\Rightarrow 12 \times x = (x - 13.20) \times 100$$

$$\Rightarrow 12x = 100x - 1320$$

$$\Rightarrow 1320 = 100x - 12x$$

$$\Rightarrow 1320 = 88x$$

$$\Rightarrow x = \frac{1320}{88}$$

$$\Rightarrow x = 15$$

Now, M.P. of the pen = ₹ 15

$$\text{Discount}\% = 5\%$$

$$\text{Discount}\% = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 5 = \frac{\text{Discount}}{15} \times 100$$

$$\Rightarrow \text{Discount} = \frac{5 \times 15}{100}$$

$$= \frac{75}{100}$$

$$= 0.75$$

And,

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\Rightarrow 0.75 = 15 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 15 - 0.75$$

$$\Rightarrow \text{S.P.} = 14.25$$

Hence, M.P. of pen = ₹ 15 and new S.P. = ₹ 14.25

Question 17

The cost price of an article is ₹ 2,400 and it is marked at 25% above the cost price. Find the profit and the profit percent, if the article is sold at 15% discount.

Answer

Given:

M.P. of the article = 25% above the C.P.

The C.P. of the article = ₹ 2,400.

$$\text{M.P.} = \text{C.P.} + 25\% \text{ of C.P.}$$

$$= 2,400 + \frac{25}{100} \times 2,400$$

$$= 2,400 + \frac{1}{4} \times 2,400$$

$$= 2,400 + \frac{2,400}{4}$$

$$= 2,400 + 600$$

$$= 3,000$$

So, M.P. of the goods = ₹ 3,000

$$\text{Discount \%} = 15\%$$

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 15 = \frac{\text{Discount}}{3000} \times 100$$

$$\Rightarrow \text{Discount} = \frac{15 \times 3,000}{100}$$

$$= \frac{45,000}{100}$$

$$= 450$$

Now,

$$\text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\Rightarrow 450 = 3,000 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 3,000 - 450$$

$$= 2,550$$

Hence, S.P. of the goods = ₹ 2,550

C.P. of the goods = ₹ 2,400

($\because$ S.P. is greater than C.P., means goods are sold at a gain.)

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \text{Gain} = 2,550 - 2,400$$

$$\Rightarrow \text{Gain} = 150$$

And

$$\text{Gain}\% = \frac{\text{Gain}}{\text{C.P.}} \times 100\%$$

$$\Rightarrow \text{Gain}\% = \frac{150}{2400} \times 100\%$$

$$= \frac{15000}{2400}\%$$

$$= 6.25\%$$

Hence, Gain = ₹ 150 and Gain percent = 6.25%

Question 18

Thirty articles are bought at ₹ 450 each. If one-third of these articles are sold at 6% loss; at what price must each of the remaining articles be sold in order to make a profit of 10% on the whole ?

Answer**Given:**

C.P. of 1 article = ₹ 450

C.P. of 30 articles = ₹ 450 x 30 = ₹ 13,500

Profit desired on the whole = 10%

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P}} \times 100$$

Putting the values, we get

$$10 = \frac{\text{Profit}}{13500} \times 100$$

$$\Rightarrow \text{Profit} = \frac{10 \times 13500}{100}$$

$$= \frac{135000}{100}$$

$$= 1350$$

As we know,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

Putting the values, we get

$$1350 = \text{S.P.} - 13500$$

$$\Rightarrow \text{S.P.} = 13500 + 1350$$

$$= 14850$$

$$\text{C.P. of } \frac{1}{3} \text{ articles} = ₹ \left(\frac{1}{3} \times 13500 \right)$$

$$= ₹ \frac{13500}{3}$$

$$= ₹ 4500$$

Loss on it = 6%

$$\text{Loss}\% = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$6 = \frac{\text{Loss}}{4500} \times 100$$

$$\Rightarrow \text{Loss} = \frac{6 \times 4500}{100}$$

$$= \frac{27000}{100}$$

$$= 270$$

As we know,

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

Putting the values, we get

$$270 = 4500 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 4500 - 270$$

$$= 4230$$

$$\text{The S.P. of the rest of the articles} = ₹ 14850 - ₹ 4230 = ₹ 10620$$

$$\text{Total articles} = 30$$

$$\text{Two - third of articles} = \frac{2}{3} \times 30 = 20$$

$$\text{The S.P. of each of the 20 articles} = ₹ \frac{10620}{20} = ₹ 531$$

The selling price of each of the article = ₹ 531.

Question 19

Find the single discount (as percent) equivalent to successive discounts of :

- (i) 80% and 80%
- (ii) 60% and 60%
- (iii) 60% and 80%

Answer

(i) Let M.P. be ₹ 100.

$$\text{1st discount \%} = 80\%$$

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 80 = \frac{\text{Discount}}{100} \times 100$$

$$\Rightarrow 80 = \frac{\text{Discount}}{100} \times 100$$

$$\Rightarrow \text{Discount} = 80$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 100 - 80$$

$$\Rightarrow \text{S.P.} = 20$$

$$\text{New M.P.} = ₹ 20$$

$$\text{2nd Discount \%} = 80\%$$

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 80 = \frac{\text{Discount}}{20} \times 100$$

$$\Rightarrow \text{Discount} = \frac{80 \times 20}{100}$$

$$\Rightarrow \text{Discount} = \frac{1600}{100}$$

$$\Rightarrow \text{Discount} = 16$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 20 - 16$$

$$\Rightarrow \text{S.P.} = 4$$

Single equivalent discount = Initial M.P. - Final S.P.

$$= 100 - 4$$

$$= 96$$

$$\text{Discount \%} = \frac{\text{Single Discount}}{\text{Initial M.P.}} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{96}{100} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{96}{\cancel{100}} \times \cancel{100}\%$$

$$\Rightarrow \text{Discount\%} = 96\%$$

Hence, single equivalent discount = 96%.

(ii) Let M.P. be ₹ 100.

$$\text{1st discount \%} = 60\%$$

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 60 = \frac{\text{Discount}}{100} \times 100$$

$$\Rightarrow 60 = \frac{\text{Discount}}{\cancel{100}} \times \cancel{100}$$

$$\Rightarrow \text{Discount} = 60$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 100 - 60$$

$$\Rightarrow \text{S.P.} = 40$$

$$\text{New M.P.} = ₹ 40$$

$$\text{2nd Discount \%} = 60\%$$

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 60 = \frac{\text{Discount}}{40} \times 100$$

$$\Rightarrow \text{Discount} = \frac{60 \times 40}{100}$$

$$\Rightarrow \text{Discount} = \frac{2400}{100}$$

$$\Rightarrow \text{Discount} = 24$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 40 - 24$$

$$\Rightarrow \text{S.P.} = 16$$

$$\text{Single equivalent discount} = \text{Initial M.P.} - \text{Final S.P.}$$

$$= 100 - 16$$

$$= 84$$

$$\text{Discount\%} = \frac{\text{Single Discount}}{\text{Initial M.P.}} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{84}{100} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{84}{\cancel{100}} \times \cancel{100}\%$$

$$\Rightarrow \text{Discount \%} = 84\%$$

Hence, single equivalent discount = 84%.

(iii) Let M.P. be ₹ 100.

$$\text{1st discount \%} = 60\%$$

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 60 = \frac{\text{Discount}}{100} \times 100$$

$$\Rightarrow 60 = \frac{\text{Discount}}{\cancel{100}} \times \cancel{100}$$

$$\Rightarrow \text{Discount} = 60$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 100 - 60$$

$$\Rightarrow \text{S.P.} = 40$$

New M.P. = ₹ 40

2nd Discount % = 80%

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 80 = \frac{\text{Discount}}{40} \times 100$$

$$\Rightarrow \text{Discount} = \frac{80 \times 40}{100}$$

$$\Rightarrow \text{Discount} = \frac{3200}{100}$$

$$\Rightarrow \text{Discount} = 32$$

And,

S.P. = M.P. - Discount

$$\Rightarrow \text{S.P.} = 40 - 32$$

$$\Rightarrow \text{S.P.} = 8$$

Single equivalent discount = Initial M.P. - Final S.P.

$$= 100 - 8$$

$$= 92$$

$$\text{Discount\%} = \frac{\text{Single Discount}}{\text{Initial M.P.}} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{92}{100} \times 100\%$$

$$\Rightarrow \text{Discount}\% = \frac{92}{100} \times 100\%$$

$$\Rightarrow \text{Discount}\% = 92\%$$

Hence, single equivalent discount = 92%.

Exercise 8(C)

Question 1(i)

Rohit buys an article marked at ₹ 600 and pays tax = 10%. The total amount paid by him is :

1. ₹ 540
2. ₹ 660
3. ₹ 590
4. ₹ 610

Answer

Selling price of the article = ₹ 600

and, Tax = 10% of ₹ 600

$$= \frac{10}{100} \times ₹ 600$$

$$= \frac{1}{10} \times ₹ 600$$

$$= ₹ \frac{600}{10}$$

$$= ₹ 60$$

Total amount to be paid by Rohit = ₹ 600 + ₹ 60

$$= ₹ 660$$

Hence, option 2 is the correct option.

Question 1(ii)

The selling price of an article is ₹ 1,000 and the tax on it is 10%. The increase of its tax if it increases to 15% is :

1. ₹ 500
2. ₹ 250
3. ₹ 50
4. ₹ 1,500

Answer

Selling price of an article = ₹ 1,000

and, Tax = 10% of ₹ 1,000

$$= \frac{10}{100} \times ₹1,000$$

$$= \frac{1}{10} \times ₹1,000$$

$$= ₹ \frac{1,000}{10}$$

$$= ₹100$$

Total amount to be paid = ₹ 1,000 + ₹ 100

$$= ₹ 1,100$$

When the selling price of the article = ₹ 1,000

and, Tax = 15% of ₹ 1,000

$$= \frac{15}{100} \times ₹1,000$$

$$= \frac{3}{20} \times ₹1,000$$

$$= ₹ \frac{3,000}{20}$$

$$= ₹150$$

Total amount to be paid = ₹ 1,000 + ₹ 150

$$= ₹ 1,150$$

The increase in tax = ₹ 1,150 - ₹ 1,100

$$= ₹ 50$$

Hence, option 3 is the correct option.

Question 1(iii)

The selling price of an article is ₹ 440 inclusive of tax at the rate of 10%, the marked price of the article is :

1. ₹ 484

2. ₹ 400

3. ₹ 450

4. ₹ 430

Answer

The selling price of an article = ₹ 440

and, Tax = 10% of marked price

Let the marked price be ₹ x .

$$x + \{10\% \text{ of } x\} = 440$$

$$\Rightarrow x + \frac{10}{100} \times x = ₹440$$

$$\Rightarrow x + \frac{1}{10} \times x = ₹440$$

$$\Rightarrow \frac{10x}{10} + \frac{x}{10} = ₹440$$

$$\Rightarrow \frac{(10x + x)}{10} = ₹440$$

$$\Rightarrow \frac{11x}{10} = ₹440$$

$$\Rightarrow x = ₹ \frac{440 \times 10}{11}$$

$$\Rightarrow x = ₹ \frac{4400}{11}$$

$$\Rightarrow x = ₹400$$

Hence, option 2 is the correct option.

Question 1(iv)

Some goods/services cost ₹ 800 and rate of GST on it is 12%. If the sales are intra-state, the total amount of bill is :

1. ₹ 896
2. ₹ 704
3. ₹ 812
4. ₹ 788

Answer

Cost of some goods/services = ₹ 800

and, GST = 12% of ₹ 800

$$= \frac{12}{100} \times ₹800$$

$$= \frac{6}{50} \times ₹800$$

$$= ₹ \frac{4,800}{50}$$

$$= ₹96$$

Total amount of bill = ₹ 800 + ₹ 96

$$= ₹ 896$$

Hence, option 1 is the correct option.

Question 1(v)

Some goods/services cost ₹ 1,500 and rate of GST on it is 8%, the total cost of the bill is :

1. ₹ 1,508

2. ₹ 1,492

3. ₹ 1,620

4. ₹ 1,380

Answer

Cost of some goods/services = ₹ 1,500

and, GST = 8% of ₹ 1,500

$$= \frac{8}{100} \times ₹1,500$$

$$= \frac{2}{25} \times ₹1,500$$

$$= ₹ \frac{3,000}{25}$$
$$= ₹120$$

Total amount of bill = ₹ 1,500 + ₹ 120

$$= ₹ 1,620$$

Hence, option 3 is the correct option.

Question 2

Ramesh paid ₹ 345.60 as tax on a purchase of ₹ 3,840. Find the rate of tax.

Answer

Given:

Sale price = ₹ 3,840

Tax paid = ₹ 345.60

Let rate of tax = x

$$\Rightarrow \frac{x}{100} \times ₹3,840 = ₹345.60$$

$$\Rightarrow x = \frac{345.60 \times 100}{3840} \Rightarrow x = \frac{34560}{3840} \Rightarrow x = 9$$

Hence, the rate of tax = 9.

Question 3

The price of a washing machine, inclusive of Tax, is ₹ 13,530/-. If the tax is 10%, find its basic (cost) price.

Answer**Given:**

Selling price of the washing machine = ₹ 13,530

and, Tax = 10% of marked price

Let the marked price be ₹ x .

$$x + 10 \% \text{ of } x = ₹13,530$$

$$\Rightarrow x + \frac{10}{100} \times x = ₹13,530$$

$$\Rightarrow x + \frac{1}{10} \times x = ₹13,530$$

$$\Rightarrow \frac{10x}{10} + \frac{x}{10} = ₹13,530$$

$$\Rightarrow \frac{(10x + x)}{10} = ₹13,530$$

$$\Rightarrow \frac{11x}{10} = ₹13,530$$

$$\Rightarrow x = ₹ \frac{13,530 \times 10}{11}$$

$$\Rightarrow x = ₹ \frac{1,35,300}{11}$$

$$\Rightarrow x = ₹12,300$$

Hence, the cost of washing machine is ₹ 12,300.

Question 4

Sarita purchases biscuits costing ₹ 158 on which the rate of tax is 6%. She also purchases some cosmetic goods costing ₹ 354 on which the rate of tax is 9%. Find the total amount to be paid by Sarita.

Answer**Given:**

Sale price of biscuits = ₹ 158

and, Tax = 6% of ₹ 158

$$= \frac{6}{100} \times ₹158$$

$$= \frac{3}{50} \times ₹158$$

$$= ₹ \frac{474}{50}$$

$$= ₹9.48$$

Total amount to be paid by Sarita for biscuits = ₹ 158 + ₹ 9.48

$$= ₹ 167.48$$

Sale price of the cosmetic goods = ₹ 354

and, Tax = 9% of ₹ 354

$$= \frac{9}{100} \times ₹354$$

$$= \frac{3,186}{100}$$

$$= ₹31.86$$

Total amount to be paid by Sartia for cosmetic goods = ₹ 354 + ₹ 31.86

$$= ₹ 385.86$$

Total amount paid by Sartia = ₹ 167.48 + ₹ 385.86

$$= ₹ 553.34$$

Question 5

The price of a T.V. set inclusive of tax of 9% is ₹ 13,407. Find its marked price. If tax is increased to 13%, how much more does the customer has to pay for the T.V. set ?

Answer

Given:

The selling price of a T.V. = ₹ 13,407

and, Tax = 9% of marked price

Let the marked price be ₹ x .

$$x + 9 \% \text{ of } x = ₹13,407$$

$$\Rightarrow x + \frac{9}{100} \times x = ₹13,407$$

$$\Rightarrow \frac{100x}{100} + \frac{9x}{100} = ₹13,407$$

$$\Rightarrow \frac{(100x + 9x)}{100} = ₹13,407$$

$$\Rightarrow \frac{109x}{100} = ₹13,407$$

$$\Rightarrow x = ₹ \frac{13,407 \times 100}{109}$$

$$\Rightarrow x = ₹ \frac{13,40,700}{109}$$

$$\Rightarrow x = ₹12,300$$

Marked price of the T.V. = ₹ 12,300

Increased Tax = 13% of marked price

$$\Rightarrow \frac{13}{100} \times ₹12,300$$

$$\Rightarrow ₹ \frac{1,59,900}{100}$$

$$\Rightarrow ₹1,599$$

Total amount paid = ₹ 12,300 + ₹ 1,599

$$= ₹ 13,899$$

Difference of selling price = ₹ 13,899 - ₹ 13,407

$$= ₹ 492$$

Hence, the cost of the T.V. is ₹ 12,300 and the customer has to pay ₹ 492 more for the T.V. when tax is increased by 13%.

Question 6

The price of an article is ₹ 8,250 which includes tax at 10%. Find how much more or less does a customer pay for the article, if the tax on the article:

(i) increases to 15%

(ii) decreases to 6%

(iii) increases by 2%

(iv) decreases by 3%.

Answer

Given:

The selling price of an article = ₹ 8,250

and, Tax = 10% of marked price

Let the marked price be ₹ x .

$$x + 10 \% \text{ of } x = ₹8,250$$

$$\Rightarrow x + \frac{10}{100} \times x = ₹8,250$$

$$\Rightarrow x + \frac{1}{10} \times x = ₹8,250$$

$$\Rightarrow \frac{10x}{10} + \frac{x}{10} = ₹8,250$$

$$\Rightarrow \frac{(10x + x)}{10} = ₹8,250$$

$$\Rightarrow \frac{11x}{10} = ₹8,250$$

$$\Rightarrow x = ₹ \frac{8,250 \times 10}{11}$$

$$\Rightarrow x = ₹ \frac{82,500}{11}$$

$$\Rightarrow x = ₹7,500$$

(i) Marked price of the article = ₹ 7,500

And, Tax = 15% of marked price

$$\Rightarrow \frac{15}{100} \times ₹7,500$$

$$\Rightarrow ₹ \frac{1,12,500}{100}$$

$$\Rightarrow ₹1,125$$

Total amount paid = ₹ 7,500 + ₹ 1,125

$$= ₹ 8,625$$

Difference in selling price = ₹ 8,625 - ₹ 8,250

$$= ₹ 375$$

Hence, the customer has to pay ₹ 375 more for the article when tax is increased by 15%.

(ii) Marked price of the article = ₹ 7,500

And, Tax = 6% of marked price

$$\Rightarrow \frac{6}{100} \times ₹7,500$$

$$\Rightarrow ₹ \frac{45,000}{100}$$

$$\Rightarrow ₹450$$

Total amount paid = ₹ 7,500 + ₹ 450

$$= ₹ 7,950$$

Difference in selling price = ₹ 8,250 - ₹ 7,950

$$= ₹ 300$$

Hence, the customer has to pay ₹ 300 less for the article when tax was decreased by 6%.

(iii) Marked price of the article = ₹ 7,500

Tax is increased by 2% means (10% + 2%) = 12%

Increased Tax = 12% of marked price

$$\Rightarrow \frac{12}{100} \times ₹7,500$$

$$\Rightarrow ₹ \frac{90,000}{100}$$

$$\Rightarrow ₹900$$

Total amount paid = ₹ 7,500 + ₹ 900

$$= ₹ 8,400$$

$$\text{Difference in selling price} = ₹ 8,400 - ₹ 8,250$$

$$= ₹ 150$$

Hence, the customer has to pay ₹ 150 more for the article when tax is increased to 2%.

$$\text{(iv) Marked price of the article} = ₹ 7,500$$

$$\text{Tax is decreased by 3\% means } (10\% - 3\%) = 7\%.$$

$$\text{Decreased Tax} = 7\% \text{ of marked price}$$

$$\Rightarrow \frac{7}{100} \times ₹ 7,500$$

$$\Rightarrow ₹ \frac{52,500}{100}$$

$$\Rightarrow ₹ 525$$

$$\text{Total amount paid} = ₹ 7,500 + ₹ 525$$

$$= ₹ 8,025$$

$$\text{Difference in selling price} = ₹ 8,250 - ₹ 8,025$$

$$= ₹ 225$$

Hence, the customer has to pay ₹ 225 less for the article when tax is decreased to 3%.

Question 7

A bicycle is available for ₹ 1,664 including tax. If the list price of the bicycle is ₹ 1,600, find

(i) the rate of Tax.

(ii) the price a customer will pay for the bicycle if the tax is increased by 6%.

Answer

(i) **Given:**

Sale price of bicycle = ₹ 1,664

List price of bicycle = ₹ 1,600

Let rate of tax = x

Tax on the sale = x of ₹ 1,600

$$\Rightarrow \frac{x}{100} \times ₹1,600$$

$$\Rightarrow \frac{1,600x}{100}$$

$$\Rightarrow 16x$$

The amount to be paid = Sale price of bicycle

$$\Rightarrow 1,600 + x \% \text{ of } 1,600 = 1,664$$

$$\Rightarrow 1,600 + 16x = 1,664$$

$$\Rightarrow 16x = 1,664 - 1,600$$

$$\Rightarrow 16x = 64$$

$$\Rightarrow x = \frac{64}{16}$$

$$\Rightarrow x = 4\%$$

Hence the rate of Tax = 4

(ii) Given:

List price of bicycle = ₹ 1,600

Tax is increased by 6%, means $(4\% + 6\%) = 10\%$ of marked price.

Increased Tax = 10% of ₹ 1,600

$$\Rightarrow \frac{10}{100} \times 1,600$$

$$\Rightarrow \frac{16,000}{100}$$

$$\Rightarrow 160$$

Total amount to be paid = ₹ 1,600 + ₹ 160

= ₹ 1,760

Hence, the price a customer will pay for the bicycle = ₹ 1,760.

Question 8

John belongs to Delhi. He buys goods worth ₹ 25,000 from a shop in Delhi. If the rate of GST is 5%, find how much money in all, will John pay for these

goods ?

Answer

Given:

List price of goods = ₹ 25,000

The rate of GST = 5% of list price

$$\Rightarrow ₹ \frac{5}{100} \times 25,000$$

$$\Rightarrow ₹ \frac{1}{20} \times 25,000$$

$$\Rightarrow ₹ \frac{25,000}{20}$$

$$\Rightarrow ₹ 1,250$$

Total amount to be paid = ₹ 25,000 + ₹ 1,250

= ₹ 26,250

John will pay ₹ 26,250 for all goods.

Question 9

Find the amount of bill for the following inter-state transaction of goods/services

: Cost of transaction = ₹ 30,000, discount = 30% and GST = 28%.

Answer

Given:

Cost of transaction = ₹ 30,000

Discount = 30% of cost of transaction

$$\Rightarrow 30 \% \text{ of ₹}30,000$$

$$\Rightarrow \frac{30}{100} \times ₹30,000$$

$$\Rightarrow 3 \times ₹3,000$$

$$\Rightarrow ₹9,000$$

Taxable cost of goods = ₹ 30,000 - ₹ 9,000 = ₹ 21,000

IGST = 28% of ₹ 21,000.

$$\Rightarrow \frac{28}{100} \times ₹21,000$$

$$\Rightarrow \frac{7}{25} \times ₹21,000$$

$$\Rightarrow ₹ \frac{147,000}{25}$$

$$\Rightarrow ₹5,880$$

Amount of bill = ₹ 21,000 + ₹ 5,880

= ₹ 26,880

Hence, amount of bill = ₹ 26,880.

Question 10

For both the following inter-state transaction of services, find the total amount of bill.

(i) Cost of services = ₹ 5,000, discount = 20% and GST = 12%

(ii) Cost of services = ₹ 12,500, discount = 40% and GST = 18%

Answer

(i) **Given:**

Cost of service = ₹ 5,000

Discount = 20% of cost of service

$$\Rightarrow 20 \% \text{ of } ₹5,000$$

$$\Rightarrow \frac{20}{100} \times ₹5,000$$

$$\Rightarrow \frac{1}{5} \times ₹5,000$$

$$\Rightarrow ₹ \frac{5,000}{5}$$

$$\Rightarrow ₹1,000$$

Taxable cost of service = ₹ 5,000 - ₹ 1,000 = ₹ 4,000

IGST = 12% of ₹ 4,000.

$$\Rightarrow \frac{12}{100} \times ₹4,000$$

$$\Rightarrow \frac{3}{25} \times ₹4,000$$

$$\Rightarrow ₹ \frac{12,000}{25}$$

$$\Rightarrow ₹480$$

$$\text{Amount of bill} = ₹ 4,000 + ₹ 480 = ₹ 4,480$$

(ii) **Given:**

$$\text{Cost of service} = ₹ 12,500$$

$$\text{Discount} = 40\% \text{ of cost of service}$$

$$\Rightarrow 40\% \text{ of } ₹12,500$$

$$\Rightarrow \frac{40}{100} \times ₹12,500$$

$$\Rightarrow \frac{10}{25} \times ₹12,500$$

$$\Rightarrow ₹ \frac{125,000}{25}$$

$$\Rightarrow ₹5,000$$

$$\text{Taxable cost of service} = ₹ 12,500 - ₹ 5,000$$

$$= ₹ 7,500$$

$$\text{IGST} = 18\% \text{ of } ₹ 7,500.$$

$$\Rightarrow \frac{18}{100} \times ₹7,500$$

$$\Rightarrow \frac{9}{50} \times ₹7,500$$

$$\Rightarrow ₹ \frac{67,500}{50}$$

$$\Rightarrow ₹1,350$$

$$\text{Amount of bill} = ₹ 7,500 + ₹ 1,350 = ₹ 8,850$$

$$\text{Total amount of bill for both the services} = ₹ 4,480 + ₹ 8,850$$

$$= ₹ 13,330$$

Hence, total amount of bill = ₹ 13,330.

Question 11

A shopkeeper in Indore sells 20 identical articles for ₹ 450 each. Find the amount of bill if he gives 20% discount and then charges GST = 28%

Answer

$$\text{Selling price of 1 article} = ₹ 450$$

$$\text{Selling price of 20 articles} = ₹ 450 \times 20$$

$$= ₹ 9,000$$

$$\text{Discount} = 20\% \text{ of } ₹ 9,000$$

$$\Rightarrow \frac{20}{100} \times ₹9,000$$

$$\Rightarrow \frac{1}{5} \times ₹9,000$$

$$\Rightarrow ₹ \frac{9,000}{5}$$

$$\Rightarrow ₹1,800$$

Taxable cost of articles = ₹ 9,000 - ₹ 1,800

$$= ₹ 7,200$$

IGST = 28% of ₹ 7,200.

$$\Rightarrow \frac{28}{100} \times ₹7,200$$

$$\Rightarrow \frac{7}{25} \times ₹7,200$$

$$\Rightarrow ₹ \frac{50,400}{25}$$

$$\Rightarrow ₹2,016$$

Amount of bill = ₹ 7,200 + ₹ 2,016

$$= ₹ 9,216$$

Hence, amount of bill = ₹ 9,216.

Question 12

A dealer in Bihar supplied goods to a dealer in Mumbai. The dealer in Mumbai buys :

(i) 40 articles for ₹ 800 each at 30% discount

(ii) 75 articles for ₹ 1,000 each at 20% discount.

If the rate of GST on the whole is 12%, find how much will the dealer at Mumbai pay to dealer in Bihar.

Answer

Selling price of 1 article = ₹ 800

Selling price of 40 articles = ₹ 800 × 40

= ₹ 3,200

Discount = 30% of ₹ 32,000

$$\Rightarrow \frac{30}{100} \times ₹32,000$$

$$\Rightarrow \frac{3}{10} \times ₹32,000$$

$$\Rightarrow ₹ \frac{96,000}{10}$$

$$\Rightarrow ₹9,600$$

Taxable cost of articles = ₹ 32,000 - ₹ 9,600

= ₹ 22,400

GST = 12% of ₹ 22,400.

$$\Rightarrow \frac{12}{100} \times ₹22,400$$

$$\Rightarrow \frac{3}{25} \times ₹22,400$$

$$\Rightarrow ₹ \frac{67,200}{25}$$

$$\Rightarrow ₹ 2,688$$

$$\text{Amount of bill} = ₹ 22,400 + ₹ 2,688$$

$$= ₹ 25,088$$

$$\text{Selling price of 1 article} = ₹ 1,000$$

$$\text{Selling price of 75 articles} = ₹ 1,000 \times 75$$

$$= ₹ 75,000$$

$$\text{Discount} = 20\% \text{ of } ₹ 75,000$$

$$\Rightarrow \frac{20}{100} \times ₹ 75,000$$

$$\Rightarrow \frac{1}{5} \times ₹ 75,000$$

$$\Rightarrow ₹ \frac{75,000}{5}$$

$$\Rightarrow ₹ 15,000$$

$$\text{Taxable cost of article} = ₹ 75,000 - ₹ 15,000$$

$$= ₹ 60,000$$

$$\text{GST} = 12\% \text{ of } ₹ 60,000.$$

$$\Rightarrow \frac{12}{100} \times ₹60,000$$

$$\Rightarrow \frac{3}{25} \times ₹60,000$$

$$\Rightarrow ₹ \frac{1,80,000}{25}$$

$$\Rightarrow ₹7,200$$

Amount of bill = ₹ 60,000 + ₹ 7,200

$$= ₹ 67,200$$

Total amount the dealer at Mumbai will pay to dealer in Bihar = ₹ 25,088 + ₹ 67,200

$$= ₹ 92,288$$

Hence, amount of bill = ₹ 92,288.

Test Yourself

Question 1(i)

A box is sold for ₹ 18 at a loss of 10%. When sold at a profit of 15%; its S.P. will be :

1. ₹ 17

2. ₹ 19

3. ₹ 21

4. ₹ 23

Answer**Given:**

S.P. = ₹ 18

Loss % = 10%

Let C.P. = ₹ x .

$$\text{Loss\%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Putting the values, we get

$$\Rightarrow 10 = \frac{\text{Loss}}{x} \times 100$$

$$\Rightarrow \text{Loss} = \frac{10 \times x}{100}$$

$$= \frac{x}{10}$$

As we know:

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{x}{10} = x - 18$$

$$\Rightarrow 18 = x - \frac{x}{10}$$

$$\Rightarrow 18 = \frac{10x}{10} - \frac{x}{10}$$

$$\Rightarrow 18 = \frac{(10x - x)}{10}$$

$$\Rightarrow 18 = \frac{9x}{10}$$

$$\Rightarrow x = \frac{18 \times 10}{9}$$

$$\Rightarrow x = \frac{180}{9}$$

$$\Rightarrow x = 20$$

When C.P. = ₹ 20

Profit % = 15 %

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow 15 = \frac{\text{Profit}}{20} \times 100$$

$$\Rightarrow \text{Profit} = \frac{15 \times 20}{100}$$

$$\Rightarrow \text{Profit} = \frac{300}{100}$$

$$\Rightarrow \text{Profit} = 3$$

And,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow 3 = \text{S.P.} - 20$$

$$\Rightarrow \text{S.P.} = ₹20 + 3$$

$$\Rightarrow \text{S.P.} = ₹23$$

Hence, option 4 is the correct option.

Question 1(ii)

C.P. of 20 articles is equal to S.P. of 16 articles. The profit or loss as percent is:

1. 25% profit
2. 25% loss
3. 20% profit
4. 20% loss

Answer

C.P. of 20 articles = S.P. of 16 articles

Let the C.P. of 20 articles be ₹ 100.

$$\text{The C.P. of 1 article} = ₹ \frac{100}{20} = ₹ 5$$

The S.P. of 16 articles = The C.P. of 20 articles = ₹ 100

$$\text{The S.P. of 1 article} = ₹ \frac{100}{16} = ₹ 6.25$$

($\because$ S.P. is greater than C.P., means articles are sold at a profit.)

Profit = S.P. - C.P.

$$= ₹ (6.25 - 5) = ₹ 1.25$$

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100\%$$

$$= \frac{1.25}{5} \times 100\%$$

$$= \frac{125}{5}\%$$

$$= 25\%$$

Hence, option 1 is the correct option.

Question 1(iii)

Some goods are sold at a discount of 20%. If the same goods are sold without discount, their price will change by :

1. 25% increase
2. 25% decrease
3. 20% increase
4. 20% decrease

Answer

Let the marked price of goods be ₹ 100.

Discount = 20% of marked price

$$\begin{aligned} &= \frac{20}{100} \times 100 \\ &= \frac{20}{\cancel{100}} \times \cancel{100} \\ &= 20 \end{aligned}$$

Selling price of goods = Marked price - Discount

$$= ₹ 100 - 20$$

$$= ₹ 80$$

Selling price without discount = marked price of the goods = ₹ 100

$$\text{Difference in the selling price} = ₹ 100 - 80$$

($\because$ Difference is positive means selling price has increased.)

$$\text{Increase in price} = \frac{\text{Difference in S.P.}}{\text{Initial M.P.}} \times 100$$

$$= \frac{20}{100} \times 100$$

$$= \frac{20}{\cancel{100}} \times \cancel{100}$$

$$= 20$$

Hence, option 3 is the correct option.

Question 1(iv)

The marked price of an article is ₹ 400. If tax on it increases from 10% to 15%, the amount of it will increase by:

1. ₹ 30
2. ₹ 40
3. ₹ 60
4. none of these

Answer

Marked price of the article = ₹ 400

and, Tax = 10% of ₹ 400

$$= \frac{10}{100} \times ₹ 400$$

$$= \frac{1}{10} \times ₹ 400$$

$$= ₹ \frac{400}{10}$$

$$= ₹40$$

Total amount to be paid = ₹ 400 + ₹ 40

$$= ₹ 440$$

When Marked price of the article = ₹ 400

and, Tax = 15% of ₹ 400

$$= \frac{15}{100} \times ₹400$$

$$= \frac{3}{20} \times ₹400$$

$$= ₹ \frac{1200}{20}$$

$$= ₹60$$

Total amount to be paid = ₹ 400 + ₹ 60

$$= ₹ 460$$

Difference in the total amount to be paid = ₹ 460 - ₹ 440

$$= ₹ 20$$

Hence, option 4 is the correct option.

Question 1(v)

If the rate of GST on an inter-state sale is 18%, the total amount for a service of ₹ 200 is

1. ₹ 36

2. ₹ 218

3. ₹ 236

4. None of these

Answer

Let the cost of services be ₹ 200

and, GST = 18% of ₹ 200

$$= \frac{18}{100} \times ₹200$$

$$= \frac{9}{50} \times ₹200$$

$$= ₹ \frac{1800}{50}$$

$$= ₹36$$

Total amount of bill = ₹ (200 + 36) = ₹ 236

Hence, option 3 is the correct option.

Question 1(vi)

Statement 1: In case of profit (i.e. if S.P. > C.P.), C.P. = S.P. x $\left(\frac{100 \times \text{Profit Percent}}{100} \right)$.

Statement 2: In case of loss (i.e. if C.P. > S.P.), S.P. = $\left(\frac{100 \times \text{C.P.}}{100 - \text{Loss Percent}} \right)$.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

When $S.P. > C.P.$, using formula

$$\text{Profit} = S.P. - C.P.$$

By formula,

$$\Rightarrow \text{Profit}\% = \frac{\text{Profit}}{C.P.} \times 100$$

$$\Rightarrow \text{Profit}\% = \frac{S.P. - C.P.}{C.P.} \times 100$$

$$\Rightarrow \text{Profit}\% = \frac{100 S.P. - 100 C.P.}{C.P.}$$

$$\Rightarrow \text{Profit}\% \times C.P. = 100 S.P. - 100 C.P.$$

$$\Rightarrow \text{Profit}\% \times C.P. + 100 C.P. = 100 S.P.$$

$$\Rightarrow C.P. (\text{Profit}\% + 100) = 100 S.P.$$

$$\Rightarrow C.P. = \frac{100 S.P.}{(\text{Profit Percent} + 100)}$$

$$\Rightarrow \text{C.P.} = \frac{100 \text{ S.P.}}{(\text{Profit Percent} + 100)}$$

So, statement 1 is false.

When $\text{C.P.} > \text{S.P.}$,

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

By formula,

$$\Rightarrow \text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

$$\Rightarrow \text{Loss \%} = \frac{\text{C.P.} - \text{S.P.}}{\text{C.P.}} \times 100$$

$$\Rightarrow \text{Loss \%} = \frac{100 \text{ C.P.} - 100 \text{ S.P.}}{\text{C.P.}}$$

$$\Rightarrow \text{Loss \%} \times \text{C.P.} = 100 \text{ C.P.} - 100 \text{ S.P.}$$

$$\Rightarrow 100 \text{ S.P.} = 100 \text{ C.P.} - \text{Loss \%} \times \text{C.P.}$$

$$\Rightarrow 100 \text{ S.P.} = 100 \text{ C.P.} - \text{Loss \%} \times \text{C.P.}$$

$$\Rightarrow 100 \text{ S.P.} = \text{C.P.} (100 - \text{Loss \%})$$

$$\Rightarrow \text{S.P.} = \frac{\text{C.P.} (100 - \text{Loss Percent})}{100}$$

So, statement 2 is false.

Hence, option 2 is the correct option.

Question 1(vii)

Assertion (A) : Two successive discounts of 10% and 5% are equal to a single discount of $14\frac{1}{2}$.

Reason (R) : Rate of discount = $\frac{\text{Discount}}{\text{M.P.}} \times 100$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Let M.P. = ₹ 100

1st discount = 10% of ₹ 100

$$= ₹ \frac{10}{100} \times 100$$

$$= ₹ 10$$

Discounted price = ₹ 100 - ₹ 10 = ₹ 90

2nd discount = 5% of ₹ 90

$$= ₹ \frac{5}{100} \times 90$$

$$= ₹ 4.50$$

Final discounted price = ₹ 90 - ₹ 4.5 = ₹ 85.50

Total discount = M.P. - Final discounted price

$$= ₹ 100 - ₹ 85.5 = ₹ 14.5$$

Since the discount of ₹ 14.5 is on ₹ 100.

∴ Single equivalent discount%

$$= \frac{\text{Total discount}}{100} \times 100 = \frac{14.5}{100} \times 100 = 14.5$$

So, assertion (A) is true.

By formula,

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

So, reason (R) is true.

Both A and R are correct, and R is not the correct explanation for A.

Hence, option 2 is the correct option.

Question 1(viii)

Assertion (A) : If selling price of an article is ₹ 400 gaining $\frac{1}{4}$ of its C.P., then gain% = 25%.

$$\text{Reason (R) : Loss} = \frac{\text{C.P.} \times \text{Loss Percent}}{100}.$$

1. Both A and R are correct, and R is the correct explanation for A.

2. Both A and R are correct, and R is not the correct explanation for A.

3. A is true, but R is false.

4. A is false, but R is true.

Answer

Let the C.P. be ₹ a .

$$\text{Gain} = \frac{1}{4} \text{ of its C.P} = \frac{1}{4} \times a$$

Using the formula,

$$\text{S.P.} = \text{Gain} + \text{C.P.}$$

$$\Rightarrow 400 = \frac{1}{4} \times a + a$$

$$\Rightarrow 400 = \frac{a + 4a}{4}$$

$$\Rightarrow 400 = \frac{5a}{4}$$

$$\Rightarrow a = \frac{4 \times 400}{5}$$

$$\Rightarrow a = \frac{1600}{5}$$

$$\Rightarrow a = ₹320$$

$$\text{Gain} = \text{S.P.} - \text{C.P.} = ₹ 400 - ₹ 320 = ₹ 80.$$

$$\text{Gain \%} = \frac{\text{Gain}}{\text{C.P.}} \times 100\%$$

$$\begin{aligned}
 &= \frac{80}{320} \times 100\% \\
 &= \frac{8000}{320}\% \\
 &= 25\%
 \end{aligned}$$

So, assertion (A) is true.

By formula,

$$\text{Loss\%} = \frac{\text{C.P.} \times \text{Loss Percent}}{100}$$

So, reason (R) is true but reason (R) does not explain assertion (A).

Hence, option 2 is the correct option.

Question 1(ix)

Assertion (A) : If M.P. is ₹ 1,200 and sales tax is 20% then amount of the bill = ₹ 1,440.

Reason (R) : M.P. is the taxable amount, hence amount of the bill =

$$M.P. \left(1 + \frac{\text{Rate of sales tax}}{100} \right)$$

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given,

$$\text{M.P.} = ₹ 1,200$$

$$\text{Sales tax} = 20\%$$

$$\text{Bill amount} = \text{M.P.} + \text{Tax amount}$$

$$= ₹ 1,200 + 20\% \text{ of } 1,200$$

$$= ₹ 1,200 + \frac{20}{100} \times 1,200$$

$$= ₹ 1,200 + 20 \times 12$$

$$= ₹ 1,200 + ₹ 240$$

$$= ₹ 1,440$$

So, assertion (A) is true.

$$\text{Bill amount} = \text{M.P.} + \text{Sale tax amount}$$

$$= \text{M.P.} + \frac{\text{Rate of sale tax}}{100} \times \text{M.P.}$$

$$= \text{M.P.} \left(1 + \frac{\text{Rate of sales tax}}{100} \right)$$

So, reason (R) is false.

Hence, option 3 is the correct option.

Question 1(x)

Assertion (A) : 12 articles are bought for one rupee and 8 of them are sold for one rupee. Then gain% = 50%.

Reason (R) : Profit % = $\left(\frac{\text{Profit}}{\text{C.P.}} \times 100\right)\%$ and Loss% = $\left(\frac{\text{Loss}}{\text{C.P.}} \times 100\right)\%$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

If 12 articles are bought for ₹1, then:

$$\text{C.P. per article} = \frac{1}{12} \text{ rupees}$$

If 8 articles are sold for ₹ 1, then:

$$\text{S.P. per article} = \frac{1}{8} \text{ rupees}$$

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\text{Profit} = \frac{1}{8} - \frac{1}{12}$$

$$= \frac{3 - 2}{24}$$

$$= \frac{1}{24}$$

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\begin{aligned}
 \text{Profit\%} &= \frac{\frac{1}{24}}{\frac{1}{12}} \times 100\% \\
 &= \frac{1 \times 12}{24 \times 1} \times 100\% \\
 &= \frac{12}{24} \times 100\% \\
 &= \frac{1}{2} \times 100\% \\
 &= 50\%
 \end{aligned}$$

So, assertion (A) is true.

$$\text{Profit \%} = \left(\frac{\text{Profit}}{\text{C.P.}} \times 100 \right)$$

These are the standard formula for finding profit% and loss%.

So, reason (R) is true but reason (R) does not clearly explains assertion (A) as there is no case of loss.

Hence, option 2 is the correct option.

Question 2

A man sold his bicycle for ₹ 405 losing one-tenth of its cost price. Find:

- (i) its cost price;
- (ii) the loss percent.

Answer

(i) **Given:**

S.P. of the bicycle = ₹ 405

Loss = one-tenth of its C.P.

Let the C.P. be ₹ x .

$$\begin{aligned}\text{Loss} &= \frac{1}{10} \times x \\ &= \frac{x}{10}\end{aligned}$$

As we know,

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{x}{10} = x - 405$$

$$\Rightarrow x - \frac{x}{10} = 405$$

$$\Rightarrow \frac{10x}{10} - \frac{x}{10} = 405$$

$$\Rightarrow \frac{(10x - x)}{10} = 405$$

$$\Rightarrow \frac{9x}{10} = 405$$

$$\Rightarrow x = \frac{405 \times 10}{9}$$

$$\Rightarrow x = \frac{4050}{9}$$

$$\Rightarrow x = 450$$

Hence, the cost price = ₹ 450.

(ii) Loss = one-tenth of the C.P.

$$= \frac{1}{10} \times 450$$

$$= \frac{450}{10}$$

$$= 45$$

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100\%$$

$$= \frac{45}{450} \times 100\%$$

$$= \frac{1}{10} \times 100\%$$

$$= 10\%$$

Hence, the loss percent = 10%.

Question 3

A man sold a radio set for ₹ 250 and gained one-ninth of its cost price. Find:

(i) its cost price;

(ii) the profit percent.

Answer

(i) **Given:**

S.P. of the radio set = ₹ 250

Gain = one-ninth of its C.P.

Let the C.P. be ₹ x .

$$\begin{aligned}\text{Gain} &= \frac{1}{9} \times x \\ &= \frac{x}{9}\end{aligned}$$

As we know,

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{9} = 250 - x$$

$$\Rightarrow \frac{x}{9} + x = 250$$

$$\Rightarrow \frac{x}{9} + \frac{9x}{9} = 250$$

$$\Rightarrow \frac{(x + 9x)}{9} = 250$$

$$\Rightarrow \frac{10x}{9} = 250$$

$$\Rightarrow x = \frac{250 \times 9}{10}$$

$$\Rightarrow x = \frac{2250}{10}$$

$$\Rightarrow x = 225$$

Hence, the cost price = ₹ 225.

(ii) Gain = one-ninth of its C.P.

$$= \frac{1}{9} \times 225$$

$$= \frac{225}{9}$$

$$= 25$$

$$\text{Gain \%} = \frac{\text{Gain}}{\text{C.P.}} \times 100\%$$

$$\Rightarrow \text{Gain \%} = \frac{25}{225} \times 100\%$$

$$= \frac{1}{9} \times 100\%$$

$$= \frac{100}{9}\%$$

$$= 11\frac{1}{9}\%$$

Hence, the gain percent = $11\frac{1}{9}$.

Question 4

Mr. Sinha sold two tape recorders for ₹ 990 each, gaining 10% on one and losing 10% on the other. Find his total loss or gain, as percent, on the whole

transaction.

Answer

Given:

S.P. of tape recorder = ₹ 990

Gain% on one tape recorder = 10%

Loss% on one tape recorder = 10%

Lets take C.P. of tape recorder be ₹ x .

As we know,

$$\text{Gain\%} = \frac{\text{Gain}}{\text{C.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Gain}}{x} \times 100$$

$$\Rightarrow \text{Gain} = \frac{10 \times x}{100}$$

$$\Rightarrow \text{Gain} = \frac{x}{10}$$

And,

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{10} = 990 - x$$

$$\Rightarrow \frac{x}{10} + x = 990$$

$$\Rightarrow \frac{x}{10} + \frac{10x}{10} = 990$$

$$\Rightarrow \frac{(x + 10x)}{10} = 990$$

$$\Rightarrow \frac{11x}{10} = 990$$

$$\Rightarrow x = \frac{990 \times 10}{11}$$

$$\Rightarrow x = \frac{9,900}{11}$$

$$\Rightarrow x = 900$$

When S.P. is ₹ 990 and gain% is 10%, then C.P. is ₹ 900.

Now, when loss% = 10%

As we know,

$$\text{Loss\%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Loss}}{x} \times 100$$

$$\Rightarrow \text{Loss} = \frac{10 \times x}{100}$$

$$\Rightarrow \text{Loss} = \frac{x}{10}$$

And,

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{x}{10} = x - 990$$

$$\Rightarrow x - \frac{x}{10} = 990$$

$$\Rightarrow \frac{10x}{10} - \frac{x}{10} = 990$$

$$\Rightarrow \frac{(10x - x)}{10} = 990$$

$$\Rightarrow \frac{9x}{10} = 990$$

$$\Rightarrow x = \frac{990 \times 10}{9}$$

$$\Rightarrow x = \frac{9,900}{9}$$

$$\Rightarrow x = 1,100$$

When S.P. is ₹ 990 and loss% is 10%, then C.P. is ₹ 1,100.

Total C.P. of both tape recorder = ₹ 900 + ₹ 1,100

= ₹ 2,000

Total S.P. of both tape recorders = ₹ 990 + ₹ 990

= ₹ 1,980

($\because$ S.P. is lesser than C.P., means article is sold at loss.)

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \text{Loss} = ₹2,000 - ₹1,980$$

$$= ₹20$$

And,

$$\text{Loss}\% = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

$$= \frac{20}{2000} \times 100\%$$

$$= \frac{1}{100} \times 100\%$$

$$= \frac{1}{\cancel{100}} \times \cancel{100}\%$$

$$= 1\%$$

Hence, the overall loss percent = 1%.

Question 5

A tape recorder is sold for ₹ 2,760 at a gain of 15% and a C.D. player is sold for ₹ 3,240 at a loss of 10%. Find

- (i) the C.P. of the tape recorder.
- (ii) the C.P. of the C.D. player.
- (iii) the total C.P. of both.
- (iv) the total S.P. of both.

(v) the gain % or the loss% on the whole.

Answer

(i) **Given:**

S.P. of tape recorder = ₹ 2,760

Gain% on one tape recorder = 15%

Let C.P. of tape recorder = ₹ x .

As we know,

$$\text{Gain\%} = \frac{\text{Gain}}{\text{C.P.}} \times 100$$

$$\Rightarrow 15 = \frac{\text{Gain}}{x} \times 100$$

$$\Rightarrow \text{Gain} = \frac{15 \times x}{100}$$

$$\Rightarrow \text{Gain} = \frac{3x}{20}$$

And,

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{3x}{20} = 2,760 - x$$

$$\Rightarrow \frac{3x}{20} + x = 2,760$$

$$\Rightarrow \frac{3x}{20} + \frac{20x}{20} = 2,760$$

$$\Rightarrow \frac{(3x + 20x)}{20} = 2,760$$

$$\Rightarrow \frac{23x}{20} = 2,760$$

$$\Rightarrow x = \frac{2,760 \times 20}{23}$$

$$\Rightarrow x = \frac{55,200}{23}$$

$$\Rightarrow x = 2,400$$

Hence, the cost price of tape recorder = ₹ 2,400.

(ii) Given:

S.P. of the CD Player = ₹ 3,240

Loss% = 10%

Let the C.P. of the CD Player be ₹ x

As we know,

$$\text{Loss\%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Loss}}{x} \times 100$$

$$\Rightarrow \text{Loss} = \frac{10 \times x}{100}$$

$$\Rightarrow \text{Loss} = \frac{x}{10}$$

And,

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{x}{10} = x - 3,240$$

$$\Rightarrow x - \frac{x}{10} = 3,240$$

$$\Rightarrow \frac{10x}{10} - \frac{x}{10} = 3,240$$

$$\Rightarrow \frac{(10x - x)}{10} = 3,240$$

$$\Rightarrow \frac{9x}{10} = 3,240$$

$$\Rightarrow x = \frac{3,240 \times 10}{9}$$

$$\Rightarrow x = \frac{32,400}{9}$$

$$\Rightarrow x = 3,600$$

Hence, the cost price of CD Player = ₹ 3,600.

(iii) Total C.P. of both = ₹ 2,400 + ₹ 3,600

= ₹ 6,000

Total C.P. of both items = ₹ 6,000

(iv) Total S.P. of both = ₹ 2,760 + ₹ 3,240

= ₹ 6,000

Total S.P. of both items = ₹ 6,000

(v) As S.P. is equal to C.P., means article is sold at neither loss nor gain.

Hence, the overall gain percent or loss percent = 0%.

Question 6

John sold an article to Peter at 20% profit and Peter sold it to Mohan at 5% loss.

If Mohan paid ₹ 912 for the article, find how much did John pay for it?

Answer

Let the cost John paid for the article be ₹ x .

John sold the article to Peter at profit = 20%

As we know

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow 20 = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{20 \times x}{100}$$

$$\Rightarrow \text{Profit} = \frac{x}{5}$$

And

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{5} = \text{S.P.} - x$$

$$\Rightarrow \text{S.P.} = \frac{x}{5} + x$$

$$\Rightarrow \text{S.P.} = \frac{x}{5} + \frac{5x}{5}$$

$$\Rightarrow \text{S.P.} = \frac{(x + 5x)}{5}$$

$$\Rightarrow \text{S.P.} = \frac{6x}{5}$$

Now, S.P. for John will be C.P. for Peter and Peter sold it to Mohan.

$$\text{C.P. of the article for Peter} = \frac{6x}{5}$$

Loss% of the article for Peter = 5%

S.P. of the article for Peter = ₹ 912

As we know

$$\text{Loss\%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

$$\Rightarrow 5 = \frac{\text{Loss}}{\frac{6x}{5}} \times 100$$

$$\Rightarrow \text{Loss} = \frac{5 \times 6x}{5 \times 100}$$

$$\Rightarrow \text{Loss} = \frac{30x}{500}$$

$$\Rightarrow \text{Loss} = \frac{3x}{50}$$

And

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{3x}{50} = \frac{6x}{5} - 912$$

$$\Rightarrow \frac{6x}{5} - \frac{3x}{50} = 912$$

$$\Rightarrow \frac{60x}{50} - \frac{3x}{50} = 912$$

$$\Rightarrow \frac{(60x - 3x)}{50} = 912$$

$$\Rightarrow \frac{57x}{50} = 912$$

$$\Rightarrow x = \frac{912 \times 50}{57}$$

$$\Rightarrow x = \frac{45,600}{57}$$

$$\Rightarrow x = 800$$

Hence, John paid ₹ 800 for the article.

Question 7

By selling an article for ₹ 1,200, Rohit loses one-fifth of its cost price. For how much should he sell it in order to gain 30% ?

Answer

Given:

S.P. of an article = ₹ 1,200

Loss = one-fifth of C.P.

Let C.P. of the article = ₹ x .

$$\begin{aligned}\text{Loss} &= \frac{1}{5} \times x \\ &= \frac{x}{5}\end{aligned}$$

As we know:

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow \frac{x}{5} = x - 1,200$$

$$\Rightarrow 1,200 = x - \frac{x}{5}$$

$$\Rightarrow 1,200 = \frac{5x}{5} - \frac{x}{5}$$

$$\Rightarrow 1,200 = \frac{(5x - x)}{5}$$

$$\Rightarrow 1,200 = \frac{4x}{5}$$

$$\Rightarrow x = \frac{1,200 \times 5}{4}$$

$$\Rightarrow x = \frac{6,000}{4}$$

$$\Rightarrow x = 1,500$$

Hence, C.P. of the article = ₹ 1,500

Gain % = 30%

$$\text{Gain \%} = \frac{\text{Gain}}{\text{C.P.}} \times 100$$

$$\Rightarrow 30 = \frac{\text{Gain}}{1,500} \times 100$$

$$\Rightarrow \text{Gain} = \frac{30 \times 1,500}{100}$$

$$\Rightarrow \text{Gain} = \frac{45,000}{100}$$

$$\Rightarrow \text{Gain} = 450$$

And,

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow 450 = \text{S.P.} - 1,500$$

$$\Rightarrow \text{S.P.} = 450 + 1,500$$

$$\Rightarrow \text{S.P.} = 1,950$$

Hence, New S.P. of the article = ₹ 1,950.

Question 8

By selling an article for ₹ 1,200; Rohit gains one-fifth of its cost price. What should be the selling price of the article when he sells it at 30% gain?

Answer

Given:

S.P. of an article = ₹ 1,200

Gain = one-fifth of C.P.

Let C.P. of the article = ₹ x .

$$\begin{aligned}\text{Gain} &= \frac{1}{5} \times x \\ &= \frac{x}{5}\end{aligned}$$

As we know:

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{5} = 1,200 - x$$

$$\Rightarrow 1,200 = x + \frac{x}{5}$$

$$\Rightarrow 1,200 = \frac{5x}{5} + \frac{x}{5}$$

$$\Rightarrow 1,200 = \frac{(5x + x)}{5}$$

$$\Rightarrow 1,200 = \frac{6x}{5}$$

$$\Rightarrow x = \frac{1,200 \times 5}{6}$$

$$\Rightarrow x = \frac{6,000}{6}$$

$$\Rightarrow x = 1,000$$

Hence, C.P. of the article = ₹ 1,000

Gain % = 30%

$$\text{Gain\%} = \frac{\text{Gain}}{\text{C.P.}} \times 100$$

$$\Rightarrow 30 = \frac{\text{Gain}}{1,000} \times 100$$

$$\Rightarrow \text{Gain} = \frac{30 \times 1,000}{100}$$

$$\Rightarrow \text{Gain} = \frac{30,000}{100}$$

$$\Rightarrow \text{Gain} = ₹300$$

And,

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow 300 = \text{S.P.} - 1,000$$

$$\Rightarrow \text{S.P.} = 300 + 1,000$$

$$\Rightarrow \text{S.P.} = ₹1,300$$

Hence, S.P. of the article = ₹ 1,300.

Question 9

25% of the cost price of an article is ₹ 600. Find its selling price when it is sold at a profit of 25%.

Answer

25% of the cost price of an article = ₹ 600

$$\frac{1}{4} \text{ of the cost price of an article} = ₹ 600$$

The cost price of an article = ₹ 600 x 4 = ₹ 2,400

Profit on the article = 25%

$$\text{Profit\%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow 25 = \frac{\text{Profit}}{2400} \times 100$$

$$\Rightarrow \text{Profit} = \frac{25 \times 2400}{100}$$

$$\Rightarrow \text{Profit} = \frac{60000}{100}$$

$$\Rightarrow \text{Profit} = 600$$

And

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow 600 = \text{S.P.} - 2,400$$

$$\Rightarrow \text{S.P.} = 600 + 2,400$$

$$\Rightarrow \text{S.P.} = 3,000$$

Hence, S.P. of the article = ₹ 3,000.

Question 10

A man sold a bicycle at 5% profit. If the cost had been 30% less and the selling price ₹ 63 less, he would have made a profit of 30%. What was the cost price of the bicycle?

Answer

Let the C.P. of the bicycle be ₹ 100.

$$\text{Profit \%} = 5\%$$

As we know

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow 5 = \frac{\text{Profit}}{100} \times 100$$

$$\Rightarrow 5 = \frac{\text{Profit}}{100} \times 100$$

$$\Rightarrow \text{Profit} = ₹5$$

And

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow 5 = \text{S.P.} - 100$$

$$\Rightarrow \text{S.P.} = 100 + 5$$

$$\Rightarrow \text{S.P.} = ₹105$$

When C.P. of the bicycle is 30 less.

$$= 100 - \frac{30}{100} \times 100$$

$$= 100 - \frac{30}{100} \times 100$$

$$= 100 - 30$$

$$= ₹70$$

$$\therefore \text{New C.P.} = ₹ 70$$

$$\text{New gain \%} = 30\%.$$

As we know,

$$\text{Gain \%} = \frac{\text{Gain}}{\text{C.P.}} \times 100$$

$$\Rightarrow 30 = \frac{\text{Gain}}{70} \times 100$$

$$\Rightarrow \text{Gain} = \frac{30 \times 70}{100}$$

$$\Rightarrow \text{Gain} = \frac{2100}{100}$$

$$\Rightarrow \text{Gain} = ₹21$$

And

$$\text{New Gain} = \text{New S.P.} - \text{New C.P.}$$

$$\Rightarrow 21 = \text{New S.P.} - 70$$

$$\Rightarrow \text{New S.P.} = 21 + 70$$

$$\Rightarrow \text{New S.P.} = ₹91$$

$$\text{Difference in Selling Price} = \text{Original S.P.} - \text{New S.P.}$$

$$= ₹ 105 - ₹ 91 = ₹ 14$$

$\therefore$ New S.P. is ₹ 14 less than original S.P.

Applying unitary method:

When sold for ₹ 14 less, the C.P. of the bicycle = ₹ 100.

$$\text{When sold for ₹ 63 less, the C.P. of the bicycle} = ₹ \frac{100}{14} \times 63 = ₹450.$$

Hence, cost price of the bicycle = ₹ 450.

Question 11

Renu sold an article at a loss of 8 percent. Had she bought it at 10% less and sold for ₹ 36 more, she would have gained 20%. Find the cost price of the article.

Answer

Let the C.P. of the article be ₹ 100.

$$\text{Loss\%} = 8\%$$

As we know

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

$$\Rightarrow 8 = \frac{\text{Loss}}{100} \times 100$$

$$\Rightarrow 8 = \frac{\text{Loss}}{100} \times \cancel{100}$$

$$\Rightarrow \text{Loss} = 8$$

And

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$\Rightarrow 8 = 100 - \text{S.P.}$$

$$\Rightarrow \text{S.P.} = 100 - 8$$

$$\Rightarrow \text{S.P.} = 92$$

When C.P. of the article is 10% less.

$$= 100 - \frac{10}{100} \times 100$$

$$= 100 - \frac{10}{100} \times \cancel{100}$$

$$= 100 - 10$$

$$= 90$$

The gain % = 20%.

As we know

$$\text{Gain\%} = \frac{\text{Gain}}{\text{C.P.}} \times 100$$

$$\Rightarrow 20 = \frac{\text{Gain}}{90} \times 100$$

$$\Rightarrow \text{Gain} = \frac{20 \times 90}{100}$$

$$\Rightarrow \text{Gain} = \frac{1800}{100}$$

$$\Rightarrow \text{Gain} = 18$$

And

$$\text{Gain} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow 18 = \text{S.P.} - 90$$

$$\Rightarrow \text{S.P.} = 18 + 90$$

$$\Rightarrow \text{S.P.} = 108$$

$$\text{Difference in Selling Price} = ₹ 108 - ₹ 92 = ₹ 16$$

Applying unitary method:

When sold for ₹ 16 more, the C.P. of the article = ₹ 100.

When sold for ₹ 36 more, the C.P. of the article = ₹ $\frac{100}{16} \times 36 = ₹ 225$.

Hence, cost price of the article = ₹ 225.

Question 12

The cost price of an article is 25% below the marked price. If the article is available at 15% discount and its cost price is ₹ 2,400, find:

- (i) its marked price
- (ii) its selling price
- (iii) the profit percent.

Answer

(i) **Given:**

C.P. = 25% below the M.P.

C.P. = ₹ 2,400

Let the M.P. be ₹ x .

$$x - \frac{25}{100} \times x = 2,400$$

$$\Rightarrow x - \frac{1}{4} \times x = 2,400$$

$$\Rightarrow \frac{4x}{4} - \frac{x}{4} = 2,400$$

$$\Rightarrow \frac{(4x - x)}{4} = 2,400$$

$$\Rightarrow \frac{3x}{4} = 2,400$$

$$\Rightarrow x = \frac{2,400 \times 4}{3}$$

$$\Rightarrow x = \frac{9,600}{3}$$

$$\Rightarrow x = ₹3,200$$

Hence, the marked price = ₹ 3,200.

(ii) M.P. of the article = ₹ 3,200

Discount = 15% of M.P.

$$\Rightarrow \frac{15}{100} \times ₹3,200$$

$$\Rightarrow \frac{3}{20} \times ₹3,200$$

$$\Rightarrow ₹ \frac{9,600}{20}$$

$$\Rightarrow ₹480$$

S.P. of the article = M.P. - Discount

$$= ₹ 3,200 - ₹480$$

$$= ₹ 2,720$$

Hence, the selling price = ₹ 2,720.

(iii) C.P. = ₹ 2,400

S.P. = ₹ 2,720

Profit = S.P. - C.P.

= ₹2,720 – ₹2,400

= ₹320

And,

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100\%$$

$$= \frac{320}{2400} \times 100\%$$

$$= \frac{32000}{2400}\%$$

$$= \frac{40}{3}\%$$

$$= 13\frac{1}{3}\%$$

Hence, the profit percent = $13\frac{1}{3}\%$.

Question 13

Find a single discount (as percent) equivalent to following successive discounts:

(i) 20% and 12%

(ii) 10%, 20% and 20%

(iii) 20%, 10% and 5%

Answer

(i) Let M.P. be ₹ 100.

1st discount % = 20%

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 20 = \frac{\text{Discount}}{100} \times 100$$

$$\Rightarrow 20 = \frac{\text{Discount}}{100} \times 100$$

$$\Rightarrow \text{Discount} = 20$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 100 - 20$$

$$\Rightarrow \text{S.P.} = 80$$

New M.P. = ₹ 80

2nd Discount % = 12%

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 12 = \frac{\text{Discount}}{80} \times 100$$

$$\Rightarrow \text{Discount} = \frac{12 \times 80}{100}$$

$$\Rightarrow \text{Discount} = \frac{960}{100}$$

$$\Rightarrow \text{Discount} = 9.6$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 80 - 9.6$$

$$\Rightarrow \text{S.P.} = 70.4$$

$$\text{Single equivalent discount} = \text{Initial M.P.} - \text{Final S.P.}$$

$$= 100 - 70.4$$

$$= 29.6$$

$$\text{Discount \%} = \frac{\text{Single Discount}}{\text{Initial M.P.}} \times 100\%$$

$$\Rightarrow \text{Discount \%} = \frac{29.6}{100} \times 100\%$$

$$\Rightarrow \text{Discount \%} = \frac{29.6}{100} \times 100\%$$

$$\Rightarrow \text{Discount \%} = 29.6\%$$

Hence, single equivalent discount = 29.6%.

(ii) Let M.P. be ₹ 100.

1st discount % = 10%

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Discount}}{100} \times 100$$

$$\Rightarrow 10 = \frac{\text{Discount}}{\cancel{100}} \times \cancel{100}$$

$$\Rightarrow \text{Discount} = 10$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 100 - 10$$

$$\Rightarrow \text{S.P.} = 90$$

New M.P. = ₹ 90

2nd Discount % = 20%

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 20 = \frac{\text{Discount}}{90} \times 100$$

$$\Rightarrow \text{Discount} = \frac{20 \times 90}{100}$$

$$\Rightarrow \text{Discount} = \frac{1800}{100}$$

$$\Rightarrow \text{Discount} = 18$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 90 - 18$$

$$\Rightarrow \text{S.P.} = 72$$

$$\text{New M.P.} = ₹ 72$$

$$\text{2nd Discount \%} = 20\%$$

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 20 = \frac{\text{Discount}}{72} \times 100$$

$$\Rightarrow \text{Discount} = \frac{20 \times 72}{100}$$

$$\Rightarrow \text{Discount} = \frac{1440}{100}$$

$$\Rightarrow \text{Discount} = 14.4$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 72 - 14.4$$

$$\Rightarrow \text{S.P.} = 57.6$$

Single equivalent discount = Initial M.P. - Final S.P.

$$= 100 - 57.6$$

$$= 42.4$$

$$\text{Discount\%} = \frac{\text{Single Discount}}{\text{Initial M.P.}} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{42.4}{100} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{42.4}{\cancel{100}} \times \cancel{100}\%$$

$$\Rightarrow \text{Discount\%} = 42.4\%$$

Hence, single equivalent discount = 42.4%.

(iii) Let M.P. be ₹ 100.

$$\text{1st discount \%} = 20\%$$

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 20 = \frac{\text{Discount}}{100} \times 100$$

$$\Rightarrow 20 = \frac{\text{Discount}}{\cancel{100}} \times \cancel{100}$$

$$\Rightarrow \text{Discount} = 20$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 100 - 20$$

$$\Rightarrow \text{S.P.} = 80$$

$$\text{New M.P.} = ₹ 80$$

$$\text{2nd Discount \%} = 10\%$$

$$\text{Discount\%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Discount}}{80} \times 100$$

$$\Rightarrow \text{Discount} = \frac{10 \times 80}{100}$$

$$\Rightarrow \text{Discount} = \frac{800}{100}$$

$$\Rightarrow \text{Discount} = 8$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 80 - 8$$

$$\Rightarrow \text{S.P.} = 72$$

$$\text{New M.P.} = ₹ 72$$

$$\text{3rd Discount \%} = 5\%$$

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 5 = \frac{\text{Discount}}{72} \times 100$$

$$\Rightarrow \text{Discount} = \frac{5 \times 72}{100}$$

$$\Rightarrow \text{Discount} = \frac{360}{100}$$

$$\Rightarrow \text{Discount} = 3.6$$

And,

$$\text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = 72 - 3.6$$

$$\Rightarrow \text{S.P.} = 68.4$$

$$\text{Single equivalent discount} = \text{Initial M.P.} - \text{Final S.P.}$$

$$= 100 - 68.4$$

$$= 31.6$$

$$\text{Discount\%} = \frac{\text{Single Discount}}{\text{Initial M.P.}} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{31.6}{100} \times 100\%$$

$$\Rightarrow \text{Discount\%} = \frac{31.6}{100} \times 100\%$$

$$\Rightarrow \text{Discount\%} = 31.6\%$$

Hence, single equivalent discount = 31.6%.

Question 14

When the rate of Tax is decreased from 9% to 6% for a coloured T.V.; Mrs Geeta will save ₹ 780 in buying this T.V. Find the list price of the T.V.

Answer

Let the list price of the T.V. be ₹ x .

Original Tax % = 9%

$\therefore$ Original Tax = 9% of ₹ x

$$\begin{aligned} &= \frac{9}{100} \times ₹x \\ &= \frac{9x}{100} \end{aligned}$$

Selling price of the T.V. = ₹ x + ₹ $\frac{9x}{100}$

$$\begin{aligned} &= ₹ \frac{100x}{100} + ₹ \frac{9x}{100} \\ &= ₹ \frac{(100x + 9x)}{100} \\ &= ₹ \frac{109x}{100} \end{aligned}$$

Reduced Tax % = 6%

$\therefore$ Reduced Tax = 6% of ₹ x

$$= \frac{6}{100} \times ₹x$$
$$= \frac{6x}{100}$$

$$\text{New Selling price of the T.V.} = ₹x + ₹\frac{6x}{100}$$

$$= ₹\frac{100x}{100} + ₹\frac{6x}{100}$$
$$= ₹\frac{(100x + 6x)}{100}$$
$$= ₹\frac{106x}{100}$$

Given, difference in the selling price = ₹ 780 [$\because$ Mrs Geeta saves ₹ 780]

$$\text{Original S.P.} - \text{New S.P.} = 780$$

$$\Rightarrow \frac{109x}{100} - \frac{106x}{100} = 780$$
$$\Rightarrow \frac{(109x - 106x)}{100} = 780$$
$$\Rightarrow \frac{3x}{100} = 780$$
$$\Rightarrow x = \frac{780 \times 100}{3}$$
$$\Rightarrow x = \frac{78,000}{3}$$
$$\Rightarrow x = 26,000$$

Hence, the list price of the T.V. = ₹ 26,000.

Question 15

A shopkeeper sells an article for ₹ 21,384 including 10% tax. However, the actual rate of Tax is 8%. Find the extra profit made by the dealer.

Answer

S.P. of the article = ₹ 21,384

Tax = 10% on the M.P.

Let the M.P. be ₹ x .

Tax = 10% on ₹ x

$$\Rightarrow \frac{10}{100} \times x$$

$$\Rightarrow \frac{1}{10} \times x$$

$$\Rightarrow \frac{x}{10}$$

S.P. of the article = M.P. + Tax on M.P.

$$\Rightarrow x + \frac{x}{10} = 21,384$$

$$\Rightarrow \frac{10x}{10} + \frac{x}{10} = 21,384$$

$$\Rightarrow \frac{(10x + x)}{10} = 21,384$$

$$\Rightarrow \frac{11x}{10} = 21,384$$

$$\Rightarrow x = \frac{21,384 \times 10}{11}$$

$$\Rightarrow x = \frac{2,13,840}{11}$$

$$\Rightarrow x = 19,440$$

$$\text{M.P.} = ₹ 19,440$$

Actual tax = 8% of the M.P.

$$\Rightarrow \frac{8}{100} \times 19,440$$

$$\Rightarrow \frac{1,55,520}{100}$$

$$\Rightarrow 1,555.20$$

Actual S.P. of the article = M.P. + Tax on M.P.

$$= ₹ 19,440 + ₹ 1,555.20$$

$$= ₹ 20,995.20$$

Profit made by the dealer = S.P. charged by dealer - Actual S.P.

$$= ₹ (21,384 - 20,995.20)$$

$$= ₹ 388.80$$

Hence, the dealer made the profit of ₹388.80

Question 16

An article is purchased for ₹ 1,792 which includes a discount of 30% and 28% GST. Find the marked price of the article.

Answer

Let Marked Price of an article be ₹ x

Discount = 30% of ₹ x

$$\Rightarrow \frac{30}{100} \times ₹x$$

$$\Rightarrow \frac{3}{10} \times ₹x$$

$$\Rightarrow ₹ \frac{3x}{10}$$

$$\text{Taxable cost of article} = ₹ x - \frac{3x}{10}$$

$$= ₹ \frac{10x}{10} - \frac{3x}{10}$$

$$= ₹ \frac{7x}{10}$$

$$\text{IGST} = 28\% \text{ of } ₹ \frac{7x}{10}$$

$$\Rightarrow ₹ \left(\frac{28}{100} \times \frac{7x}{10} \right)$$

$$\Rightarrow ₹ \left(\frac{7}{25} \times \frac{7x}{10} \right)$$

$$\Rightarrow ₹ \frac{49x}{250}$$

Amount of bill = Taxable Cost + Tax = 1,792

$$\Rightarrow ₹ \frac{7x}{10} + ₹ \frac{49x}{250} = 1,792$$

$$\Rightarrow ₹ \frac{175x}{250} + ₹ \frac{49x}{250} = 1,792$$

$$\Rightarrow ₹ \frac{(175x + 49x)}{250} = 1,792$$

$$\Rightarrow ₹ \frac{224x}{250} = 1,792$$

$$\Rightarrow x = ₹ \left(\frac{1,792 \times 250}{224} \right)$$

$$\Rightarrow x = ₹ \left(\frac{4,48,000}{224} \right)$$

$$\Rightarrow x = ₹2,000$$

Hence, the marked price of the article = ₹ 2,000.

Question 17

The list price of a bicycle is ₹ 4,000. It is sold at a discount of 18%. If GST is 18%, find the selling price including GST.

Answer

The list price of a bicycle = ₹ 4,000

Discount = 18% of ₹ 4,000

$$\Rightarrow \frac{18}{100} \times ₹4,000$$

$$\Rightarrow \frac{9}{50} \times ₹4,000$$

$$\Rightarrow ₹ \frac{36,000}{50}$$

$$\Rightarrow ₹720$$

Taxable cost of article = ₹ (4,000 - 720)

$$= ₹ 3,280$$

IGST = 18% of ₹ 3,280

$$\Rightarrow \frac{18}{100} \times ₹3,280$$

$$\Rightarrow \frac{9}{50} \times ₹3,280$$

$$\Rightarrow ₹ \frac{29,520}{50}$$

$$\Rightarrow ₹590.40$$

Amount of bill = ₹ 3,280 + ₹ 590.40 = ₹ 3,870.40

Hence, the selling price of the bicycle = ₹ 3,870.40

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